

The Price of Fairness in Active Learning: Fundamental Limits and Optimal Label Acquisition

Anonymous Author(s)

Abstract

When should a fair classifier’s labeling budget depend on which fairness notion it must satisfy? We give a sharp answer through the first information-theoretic lower bounds for fair active learning. Consider learning a binary classifier h of Vapnik–Chervonenkis (VC) dimension d over k protected groups with classification error at most ϵ and fairness violation at most γ .

For *Demographic Parity* (DP), fairness verification is label-free: positive prediction rates can be checked from unlabeled data alone, so label complexity stays at $\Omega(d/\epsilon^2)$, identical to unconstrained learning. *Equal Opportunity* (EO) is fundamentally harder: conditioning on the true label $Y \in \{0, 1\}$ to evaluate group-wise true positive rates creates an irreducible labeling bottleneck. We prove that any active learner satisfying EO requires $\Omega(d/\epsilon^2 + k/(\gamma^2 p_+))$ labels, where $p_+ = \Pr[Y=1]$. The fairness term reflects a simple accounting: each of the k groups needs $\Omega(1/(\gamma^2 p_+))$ queries to accumulate enough labeled positives for TPR estimation, and no query strategy can avoid this. In the realizable setting, this additive cost collapses the exponential query savings of classical active learning to polynomial rates. Extending to *Equalized Odds* (EqOdds), which also constrains false positive rates, yields a complete hierarchy: DP (zero additional label cost) < EO (+ $k/(\gamma^2 p_+)$) < EqOdds (+ $k/(\gamma^2 p_+ p_-)$, $p_- = 1 - p_+$).

We present matching algorithms—CONSTRAINEDERM-DP for DP and FAIRSTRATIFIED for EO/EqOdds—with label complexity $O(d\epsilon^{-2} \log \delta^{-1} + k\gamma^{-2} p_+^{-1} \log(k/\delta))$ (failure probability δ), tight up to logarithmic factors. Experiments on Folktables ($k=9$), COMPAS ($k=6$), and Adult ($k=5$) confirm the predicted scaling and yield 36–39% label savings over fairness-agnostic baselines; at $p_+=0.01$ (e.g., fraud detection), EO requires $\sim 90\times$ more labels than DP.

1 Introduction

Active learning lets a learner choose which instances to label, rather than passively receiving a random training set [24, 60]. In domains where expert annotations are expensive—medical diagnosis, criminal justice risk assessment, content moderation—this can cut labeling budgets by orders of magnitude while preserving predictive accuracy [8, 27]. Yet these are also the domains where algorithmic fairness matters most. A recidivism predictor that performs well on average but systematically fails for certain demographic groups does real harm [5, 22].

Fair classification is now a mature field, with formal definitions—Demographic Parity (DP) [48], Equal Opportunity (EO) [37], Equalized Odds (EqOdds) [68]—and a rich toolkit of constrained optimization [3, 33, 75], post-processing [7, 48], and subgroup auditing methods [51, 53]. Nearly all of this work, however, takes a fully labeled dataset as given. How to *acquire* labels in a way that supports fair training remains largely unexplored [23].

The difficulty/challenge is that active learning and fairness pull in opposite directions. An active learner seeking maximum accuracy

concentrates queries near the decision boundary, which may leave minority groups with too few labeled examples to evaluate—let alone enforce—group-level constraints [4, 20, 31]. *What, then, is the irreducible label cost of fairness in active learning?*

Setup and notation. Let (X, Y, A) be an instance drawn from a fixed but unknown distribution, where $X \in \mathcal{X}$ is the feature vector, $Y \in \{0, 1\}$ the binary label, and $A \in \{1, \dots, k\}$ a sensitive attribute identifying one of k protected groups ($\Pr[\cdot]$ denotes probability throughout). For example, in income prediction, X encodes age, education, and occupation; $Y=1$ indicates income $>\$50K$; $A \in \{1, \dots, 9\}$ indexes race ($k=9$); the positive rate $p_+ = \Pr[Y=1] \approx 0.25$ means roughly one in four individuals earns above the threshold; and the tolerance $\gamma=0.05$ requires that the true positive rate $\text{TPR}_a(h) = \Pr[h(X)=1 \mid Y=1, A=a]$ differs by at most 0.05 across all groups, so the classifier cannot systematically miss high earners in any racial group. A learner selects a classifier $h: \mathcal{X} \rightarrow \{0, 1\}$ from a hypothesis class $\mathcal{H}$ of Vapnik–Chervonenkis (VC) dimension d , aiming for classification error $\Pr[h(X) \neq Y] \leq \epsilon$ while satisfying a fairness constraint to tolerance $\gamma > 0$. We study the *label complexity*—the minimum number of label queries any active learner must make—as a function of d, k, ϵ , and γ , and prove that the answer hinges on which fairness notion is adopted.

DP is free; EO is not. We notice that *Demographic Parity* (DP) [37] requires equal positive prediction rates $r_a(h) = \Pr[h(X)=1 \mid A=a]$ across all groups $a \in \{1, \dots, k\}$. Because $r_a(h)$ depends only on the classifier’s outputs and group membership—both observable without labels—DP can be verified using unlabeled data alone. The label complexity for learning a DP-fair classifier therefore remains $\Omega(d/\epsilon^2)$, the same as unconstrained learning.

Equal Opportunity (EO) [48] is fundamentally different though. It requires equal true positive rates (TPRs) across groups, defined as $\text{TPR}_a(h) = \Pr[h(X)=1 \mid Y=1, A=a]$; the EO gap is $\max_{a,b} |\text{TPR}_a(h) - \text{TPR}_b(h)|$, and we require it to be at most γ . The key difference from DP is the conditioning on $Y=1$: to estimate TPRs, the learner must see labeled positives from every group, and with positive-class rate $p_+ = \Pr[Y=1]$, each query yields a useful positive with probability only p_+ . This creates an information-theoretic bottleneck.

We prove that any active learner achieving error ϵ and EO gap at most γ requires at least

$$\Omega\left(\frac{d}{\epsilon^2} + \frac{k}{\gamma^2 p_+}\right)$$

label queries (Theorem 1). The first term is the standard learning cost [39, 61]; the second is an irreducible fairness tax—one group’s worth of TPR estimation costs $1/(\gamma^2 p_+)$, and there are k groups. No query strategy can avoid this cost. In the realizable setting (when the true labeling function belongs to the hypothesis class), unconstrained active learning can achieve error ϵ with $O(\log(1/\epsilon))$ queries [6, 10, 44]; the additive $k/(\gamma^2 p_+)$ term wipes out this exponential speedup entirely (Theorem 2).

A *complete label-complexity hierarchy*. Extending to *Equalized Odds* (EqOdds) [48], which constrains both TPRs and false positive rates (FPRs), the fairness cost grows further because negative examples (at rate $p_- = 1 - p_+$) are now also needed. We establish the complete label-complexity hierarchy (Corollary 1):

$$\underbrace{\text{DP}}_{\text{zero extra label cost}} < \underbrace{\text{EO}}_{+k/(\gamma^2 p_+)} < \underbrace{\text{EqOdds}}_{+k/(\gamma^2 p_+ p_-)}.$$

Matching algorithms. We complement the lower bounds with algorithms that are tight up to logarithmic factors (Theorem 3). For DP, a constrained empirical risk minimization (ERM) approach evaluates fairness entirely on unlabeled data. For EO and EqOdds, FAIRSTRATIFIED, a two-phase algorithm, achieves label complexity

$$O\left(\frac{d}{\varepsilon^2} \log \frac{1}{\delta} + \frac{k}{\gamma^2 p_+} \log \frac{k}{\delta}\right),$$

where δ is the failure probability.

Experiments on synthetic data and real-world benchmarks (Folktables, COMPAS, Adult) confirm the predicted scaling laws and yield 36–39% label savings over fairness-agnostic baselines.

2 Related Work

Label complexity of active learning. Passive learning of a hypothesis class with VC dimension d to error ε requires $\Theta(d/\varepsilon^2)$ labeled examples [39, 61, 67]. Disagreement-based active learning can reduce this to $O(\theta d \log(1/\varepsilon))$, where θ is the disagreement coefficient, bounded for linear classifiers under log-concave distributions [10, 47]. Dasgupta [27, 28] characterized when such speedups are achievable; further work analyzed Perceptron-based [30] and margin-based [9] variants. Hanneke [45] established minimax-optimal rates and gave a general passive-to-active reduction [46]. Beyond realizability, importance-weighted agnostic methods [8, 14, 50, 54], adversarial-noise robustness [6], observational settings [70], and abstention [78] extend the theory. On the lower-bound side, minimax techniques [2, 18, 26, 35, 66, 71, 73] underpin the framework we adapt; adaptive strategies extend it to structured settings [21, 43]. In practice, Query-by-Committee [42] and uncertainty sampling [24, 60] concentrate queries near the decision boundary, potentially neglecting groups whose mass lies elsewhere.

Group fairness. DP asks that positive prediction rates be equal across groups [16, 41, 52]. Hardt et al. [48] introduced EO and EqOdds; this DP–EO–EqOdds progression directly shapes our complexity hierarchy. Individual fairness [37], metric fairness [72], and the taxonomy of Verma and Rubin [68] offer complementary perspectives. These notions conflict with each other and with accuracy. Kleinberg et al. [55] and Chouldechova [22] showed calibration and balance are jointly satisfiable only when base rates coincide, motivating our γ -tolerance formulation. The accuracy cost of fairness has been studied under DP constraints [25, 59], fair representations [77], and non-discriminatory prediction [69]. Liu et al. [56, 57] showed static constraints can cause long-term harm; Dwork and Ilvento [38] studied fairness degradation under composition. Algorithmically, Agarwal et al. [3] reduced fair classification to cost-sensitive learning; Zafar et al. [74, 75] formulated convex proxy constraints; and Celis et al. [19] gave meta-algorithms with provable guarantees. Representation learning [36, 76], subgroup

auditing [53], and multicalibration [51] broaden the toolbox (surveys [58], [12], [64]). Throughout, labeled data is taken as given.

Fair data acquisition. A line of work questions that assumption. Chen et al. [20] showed collection bias drives discrimination even when algorithms enforce fairness. Blum and Stangl [15] analyzed recovery from biased data, Benade et al. [13] studied fair division under partial information, and Chouldechova and Roth [23] identified a theoretical foundation for fair data collection as an open problem. Methods range from group-balanced uncertainty sampling [4] and mixed-initiative query design [63], to bandit formulations [1, 65] and clustering-based approaches [40]; Camilleri et al. [17] provide finite-sample bounds in low-data regimes. Shen et al. [62] proved label complexity bounds for *metric-fair* active learning, a setting orthogonal to the group-fairness notions studied here.

What is missing from this landscape is an information-theoretic understanding of *why* group-fair active learning is harder than its unconstrained counterpart, and by exactly how much. We fill this gap: our lower bounds identify the precise structural mechanism—conditioning on $Y=1$ for EO, and on both $Y=1$ and $Y=0$ for EqOdds—that creates an irreducible label cost absent from DP, and our algorithms show that this cost is tight.

3 Preliminaries

Let $\mathcal{D}$ denote the joint distribution over $(X, Y, A) \in X \times \{0, 1\} \times [k]$, where $[k] := \{1, \dots, k\}$. Write $\pi_a = \Pr[A=a]$ for the proportion of group a , $p_a = \Pr[Y=1 \mid A=a]$ for the group-specific positive rate, and $p_- = 1 - p_+$ for the marginal negative-class rate (where $p_+ = \Pr[Y=1] = \sum_{a=1}^k \pi_a p_a$ is the marginal positive rate from Section 1). The classification error of a hypothesis $h \in \mathcal{H}$ (recall $\mathcal{H}$ has VC dimension d) is $\text{err}(h) = \Pr_{(X,Y) \sim \mathcal{D}}[h(X) \neq Y]$; we write $\text{opt}_{\mathcal{H}} = \inf_{h \in \mathcal{H}} \text{err}(h)$ for the best-in-class error.

For group $a \in [k]$, define the *positive prediction rate* $r_a(h) = \Pr[h(X)=1 \mid A=a]$ and the *false positive rate* $\text{FPR}_a(h) = \Pr[h(X)=1 \mid Y=0, A=a]$; the true positive rate $\text{TPR}_a(h)$ was defined in Section 1.

DEFINITION 1 (FAIRNESS GAPS). *The DP, EO, and EqOdds gaps of a classifier h are*

$$\text{DP}(h) = \max_{a, a' \in [k]} |r_a(h) - r_{a'}(h)|,$$

$$\text{EO}(h) = \max_{a, a' \in [k]} |\text{TPR}_a(h) - \text{TPR}_{a'}(h)|,$$

$$\text{EqOdds}(h) = \max_{a, a' \in [k]} \{\text{EO}(h), \max |\text{FPR}_a(h) - \text{FPR}_{a'}(h)|\}.$$

These three notions form a hierarchy of increasing stringency. DP constrains predictions regardless of true labels; EO constrains errors on positive instances ($Y=1$); EqOdds additionally constrains errors on negative instances ($Y=0$). The implication $\text{EqOdds}(h) \leq \gamma$ entails $\text{EO}(h) \leq \gamma$, but neither implies bounded $\text{DP}(h)$.

The rate $r_a(h)$ depends only on classifier predictions and group membership, both observable without labels. In contrast, $\text{TPR}_a(h)$ conditions on $Y=1$ and $\text{FPR}_a(h)$ conditions on $Y=0$, making them intrinsically label-dependent. This distinction drives the complexity separation between DP and EO/EqOdds formalized in Section 4.

An active learner has access to a pool of unlabeled instances drawn i.i.d. from the marginal of X under $\mathcal{D}$, with group memberships A observable. At each step the learner selects an instance and queries an oracle for its label Y ; after n queries it outputs $h \in \mathcal{H}$.

DEFINITION 2 (PAC-FAIR ACTIVE LEARNER). *An algorithm is an $(\epsilon, \gamma, \delta)$ -PAC fair active learner for fairness notion $\mathcal{F} \in \{\text{DP}, \text{EO}, \text{EqOdds}\}$ if, for any distribution $\mathcal{D}$, it outputs h satisfying*

$$\Pr[\text{err}(h) \leq \epsilon \text{ and } \mathcal{F}(h) \leq \gamma] \geq 1 - \delta,$$

where $\epsilon \in (0, 1)$ is the error tolerance, $\gamma \in (0, 1)$ the fairness tolerance, $\delta \in (0, 1)$ the failure probability, and the probability is over sampling and algorithmic randomness.

DEFINITION 3 (FAIR LABEL COMPLEXITY). *The $(\epsilon, \gamma, \delta)$ -fair label complexity for fairness notion $\mathcal{F}$ is the minimum number of label queries n such that an $(\epsilon, \gamma, \delta)$ -PAC fair active learner achieving the guarantee of Definition 2 exists using at most n queries in expectation.*

Without fairness constraints, passive learning requires $\Theta(d/\epsilon^2)$ labels [61]; active learning reduces this to $O(\theta d \log(1/\epsilon))$ when the disagreement coefficient θ is bounded [47]. Section 4 characterizes how fairness constraints modify these baselines.

4 Information-Theoretic Lower Bounds

We now establish that the label complexity of fair active learning depends structurally on the fairness notion adopted.

THEOREM 1 (MAIN LOWER BOUND). *Let $\mathcal{H}$ be a hypothesis class with VC dimension $d \geq 2$, $k \geq 2$ protected groups, and $0 < p_+ \leq 1/2$. For any $\epsilon \in (0, p_+/4)$, $\gamma \in (0, 1/8]$, $\delta \in (0, 1/8)$, and a universal constant $c > 0$: **(DP)** Any $(\epsilon, \gamma, \delta)$ -PAC fair active learner for DP requires $n \geq c \cdot d/\epsilon^2$ label queries. Fairness verification costs $\Omega(k/\gamma^2)$ unlabeled samples only. **(EO)** Any $(\epsilon, \gamma, \delta)$ -PAC fair active learner for EO requires*

$$n \geq c \cdot \left(\frac{d}{\epsilon^2} + \frac{k}{\gamma^2 p_+} \right).$$

(EqOdds) Any $(\epsilon, \gamma, \delta)$ -PAC fair active learner for EqOdds requires

$$n \geq c \cdot \left(\frac{d}{\epsilon^2} + \frac{k}{\gamma^2 p_+ p_-} \right),$$

where $p_- = 1 - p_+$.

The d/ϵ^2 term is the classical sample complexity for accuracy, unavoidable even without fairness constraints. The fairness terms arise from verifying group-conditional rates and are *additive*: when one term dominates, the other is essentially free. We prove each component below.

4.1 Accuracy Lower Bound

LEMMA 1 (ACCURACY LOWER BOUND). *Let $\mathcal{H}$ have VC dimension $d \geq 2$. Any algorithm achieving $\text{err}(h) \leq \epsilon$ with probability $\geq 2/3$ requires $\Omega(d/\epsilon^2)$ labeled examples, even with adaptive queries.*

PROOF SKETCH. Note that the bound $\Omega(d/\epsilon^2)$ is classical [39, 61]; we include a sketch adapted to active learning. Let $\{x_1, \dots, x_d\}$ be shattered by $\mathcal{H}$. Introduce a background point x_0 and place mass $1 - d\epsilon^2$ on x_0 and mass ϵ^2 on each x_j . Set $\Pr_s[Y=1 \mid X=x_0] = p_+$ and $\Pr_s[Y=1 \mid X=x_j] = p_+ + s_j/2$ for $s \in \{-1, +1\}^d$. Since $p_+ \leq 1/2$, these are valid probabilities and labels at each signal point are nearly deterministic; the Bayes error is $\min(p_+, 1-p_+)(1-d\epsilon^2) < 1/2$, so classifiers with error $\leq \epsilon$ exist. The Bayes-optimal prediction at x_j is $1[s_j=+1]$, and misidentifying s_j costs excess error ϵ^2 . Because each x_j has mass ϵ^2 , it is queried $\Theta(n\epsilon^2/d)$ times in expectation

from n total queries, and identifying $\Omega(d)$ coordinates requires $\Omega(1)$ queries per coordinate. This gives $n = \Omega(d/\epsilon^2)$. Formally, the per-sample KL divergence between any two hypotheses differing in one coordinate is $\Theta(\epsilon^2)$, and Assouad's lemma over the d -dimensional hypercube yields the stated bound. $\square$

4.2 DP Verification Is Label-Free

LEMMA 2 (DP VERIFICATION). *For any $h \in \mathcal{H}$ and group $a \in [k]$, the positive prediction rate $r_a(h) = \Pr[h(X)=1 \mid A=a]$ can be estimated to precision $\gamma/8$ using $O((d \log(d/\gamma) + \log(k/\delta))/\gamma^2)$ unlabeled samples per group.*

PROOF SKETCH. The empirical rate $\hat{r}_a(h) = m^{-1} \sum_i 1[h(x_i)=1]$ requires only evaluating h on unlabeled instances. Since the function class $\{x \mapsto 1[h(x)=1] : h \in \mathcal{H}\}$ has VC dimension d , standard uniform convergence (see Appendix B.2) yields

$$\sup_{h \in \mathcal{H}} |\hat{r}_a(h) - r_a(h)| \leq c_0 \sqrt{\frac{d \log(m/d) + \log(k/\delta)}{m}}$$

with probability $\geq 1 - \delta/(2k)$. Setting this bound to $\gamma/8$ and solving for m gives the stated sample complexity. $\square$

This establishes the DP part of Theorem 1: label complexity is $\Omega(d/\epsilon^2)$ from accuracy alone, with fairness verification absorbed entirely by the unlabeled pool.

4.3 Conditional Fairness Lower Bounds

For EO and EqOdds, labels become essential because TPR and FPR condition on the true label Y . The challenge is to construct distributions where learning is easy yet fairness verification demands many labels.

LEMMA 3 (TPR VERIFICATION). *Let $0 < p_+ \leq 1/2$ and $\gamma \leq 1/8$. Any algorithm that achieves $\text{err}(h) \leq \epsilon$ and certifies EO(h) $\leq \gamma$ with probability $\geq 2/3$ requires $\Omega(k/(\gamma^2 p_+))$ labeled examples.*

PROOF SKETCH. Insight: even when accuracy is trivially achievable, verifying equal opportunity requires many labels because the learner must estimate group-conditional TPRs to precision γ , and each query yields a useful positive with probability only p_+ .

Consider threshold classifiers on $[0, 1]$ with $X \sim \text{Uniform}[0, 1]$, equal group proportions $\pi_a = 1/k$, and threshold $t^* = 1 - p_+$. Set a small noise parameter $\eta = \epsilon/4$ and define

$$\Pr[Y=1 \mid X, A=a] = \begin{cases} 1 - \eta + \beta_a & \text{if } X \geq t^*, \\ \eta - \frac{p_+}{1-p_+} \beta_a & \text{if } X < t^*, \end{cases}$$

with group-specific perturbations β_a . The marginal is preserved: $\Pr[Y=1 \mid A=a] = p_+$ for all β_a . Because labels are nearly deterministic ($\eta \ll 1$), the classifier h_{t^*} achieves $\text{err}(h_{t^*}) = \eta = \epsilon/4 < \epsilon$, so the learning task is feasible. The group-level TPR is $\text{TPR}_a(h_{t^*}) = 1 - \eta + \beta_a$.

We test two hypotheses. Under H_0 , $\beta_a = 0$ for all a , giving $\text{TPR}_a = 1 - \eta$ uniformly and $\text{EO} = 0$. Under H_1 , a uniformly random group a^* receives $\beta_{a^*} = 2\gamma$ (others stay at 0), raising TPR_{a^*} to $1 - \eta + 2\gamma$ and giving $\text{EO} = 2\gamma > \gamma$; any PAC-fair learner must therefore distinguish the two cases. Because $\Pr[Y=1 \mid A=a] = p_+$ is identical under both hypotheses and the per-query KL divergence for group a^* is $O(\gamma^2 p_+)$ (the $1/p_+$ factor arises because only the

fraction p_+ of queries that land on positives carry information about the TPR perturbation), Pinsker's inequality and the randomization over a^* yield $n \geq \Omega(k/(\gamma^2 p_+))$. Full details are in Appendix B.3. $\square$

An entirely symmetric argument—swapping the roles of $Y=0$ and $Y=1$ and replacing p_+ with $p_- = 1 - p_+$ —shows that certifying equal FPRs requires $\Omega(k/(\gamma^2 p_-))$ labels (Lemma 5 in Appendix B.4).

PROOF OF THEOREM 1, EO AND EQODDS. For EO, taking the maximum of the accuracy lower bound (Lemma 1) and the TPR verification lower bound (Lemma 3) gives $\Omega(d/\varepsilon^2 + k/(\gamma^2 p_+))$. For EqOdds, EO and FPR constraints are simultaneously required; combining Lemmas 3 and 5 yields

$$\max\left(\frac{k}{\gamma^2 p_+}, \frac{k}{\gamma^2 p_-}\right) \geq \frac{k}{2\gamma^2 p_+ p_-},$$

where the inequality uses $\max(1/p_+, 1/p_-) \geq (1/p_+ + 1/p_-)/2$. Adding the accuracy term completes the proof. $\square$

4.4 Fairness Complexity Hierarchy

The lower bounds, together with the matching upper bounds in Section 5, establish a clean hierarchy of fairness verification costs.

COROLLARY 1 (FAIRNESS COMPLEXITY HIERARCHY). Let $L_{\mathcal{F}}$ denote the fairness-specific label cost (excluding the d/ε^2 accuracy baseline). Then

$$\underbrace{L_{\text{DP}} = 0}_{\text{label-free}} < \underbrace{L_{\text{EO}} = \Theta(k/(\gamma^2 p_+))}_{\text{TPR verification}} \leq \underbrace{L_{\text{EqOdds}} = \Theta(k/(\gamma^2 p_+ p_-))}_{\text{TPR + FPR}}.$$

The 1st inequality is strict for all $p_+ \in (0, 1)$; the 2nd is strict for all $p_+ \in (0, 1)$ as well, since $1/p_+ < 1/(p_+ p_-)$ holds whenever $p_- < 1$.

Table 1 instantiates these costs for two representative base rates.

Table 1: Fairness verification label cost (excluding the d/ε^2 accuracy baseline).

Notion	Label cost	$p_+=0.5$	$p_+=0.1$
DP	0	0	0
EO	$k/(\gamma^2 p_+)$	$2k/\gamma^2$	$10k/\gamma^2$
EqOdds	$k/(\gamma^2 p_+ p_-)$	$4k/\gamma^2$	$\approx 11.1k/\gamma^2$

4.5 Separation Theorem

The hierarchy above implies a qualitative separation between constrained and unconstrained active learning.

THEOREM 2 (SEPARATION). In the realizable setting, there exist hypothesis classes for which unconstrained active learning achieves error ε with $O(\theta d \log(1/\varepsilon))$ labels (where θ is the disagreement coefficient), yet any EO-constrained algorithm requires $\Omega(k/(\gamma^2 p_+))$ labels. Conditional fairness therefore eliminates exponential query savings whenever $k/(\gamma^2 p_+) \gg \theta d \log(1/\varepsilon)$.

PROOF SKETCH. The unconstrained bound follows from standard disagreement-based analysis for threshold classifiers (or half-spaces under log-concave distributions), which have bounded θ . The constrained lower bound $\Omega(k/(\gamma^2 p_+))$ is a direct consequence of Lemma 3, which is a minimax statement applying to all algorithms

regardless of whether a realizable distribution is promised. A complementary realizable-only argument (setting $\Pr[Y=1 \mid X \geq t^*] = 1$, $\Pr[Y=1 \mid X < t^*] = 0$ under H_0 and introducing noise $\Pr[Y=1 \mid X < t^*, A=a^*] = 2\gamma$ under H_1) yields a weaker bound $\Omega(k/\gamma)$ due to the degenerate KL divergence $D(\text{Ber}(0) \parallel \text{Ber}(2\gamma)) = O(\gamma)$, which nonetheless suffices for separation since $k/\gamma \gg \theta d \log(1/\varepsilon)$ in the regime of interest. $\square$

DP-fair learning retains exponential speedups because DP verification is label-free, reinforcing the structural dichotomy at the heart of our theory.

5 Algorithms and Upper Bounds

We now present algorithms whose label complexity matches the lower bounds of Section 4 up to logarithmic factors.

5.1 DP-Fair Learning via Constrained ERM

Because $r_a(h) = \Pr[h(X)=1 \mid A=a]$ is computable from unlabeled data, DP can be enforced during optimization at zero label cost for fairness (Algorithm 1).

Algorithm 1: CONSTRAINEDERM-DP

Require: Labeled pool, unlabeled pool with group membership, $\mathcal{H}$, ε , γ , δ

Phase 1 (Labels for accuracy):

Draw $n_L = O(d\varepsilon^{-2} \log \delta^{-1})$ i.i.d. labeled samples S_L .

Phase 2 (Unlabeled for fairness):

For each group $a \in [k]$, draw $m_a = O((d \log(d/\gamma) + \log(k/\delta))/\gamma^2)$ unlabeled samples $\mathcal{U}_a$ and compute $\hat{r}_a(h)$ for all $h \in \mathcal{H}$.

Solve:

$\hat{h} \leftarrow \arg \min_{h: \widehat{\text{DP}}(h) \leq \gamma/2} \widehat{\text{err}}(h; S_L)$.

return $\hat{h}$

5.2 EO/EqOdds-Fair Learning via Stratified Sampling

For conditional fairness, sufficient positive (and, for EqOdds, negative) labeled instances per group must be collected before fairness can be evaluated. FAIRSTRATIFIED (Algorithm 2) achieves this through a two-phase design.

Two design choices are critical. First, S_{cstr} and S_{val} are disjoint subsets of Phase 1 data, ensuring that $\hat{h}$ (which depends on S_{cstr} through the feasible set) is independent of S_{val} ; this independence is required for the validation step's concentration guarantee. Second, Phase 1 uses a negative-binomial stopping rule that produces samples with a biased class distribution, making them unsuitable for ERM; however, the conditional distribution $X \mid Y=1, A=a$ is preserved, which is all TPR estimation requires. Phase 2 therefore draws a separate i.i.d. sample for accuracy training.

5.3 Theoretical Guarantees

ASSUMPTION 1 (PAC-FAIR REALIZABILITY). There exists $h^* \in \mathcal{H}$ with $\text{err}(h^*) \leq \varepsilon/4$ and $\mathcal{F}(h^*) \leq \gamma/8$. (In particular, this requires $\text{opt}_{\mathcal{H}} \leq \varepsilon/4$.)

THEOREM 3 (UPPER BOUNDS). Under Assumption 1, for $\varepsilon, \gamma \in (0, 1/4)$ and $\delta \in (0, 1)$: **(DP)** CONSTRAINEDERM-DP achieves $(\varepsilon, \gamma, \delta)$ -PAC fairness with label complexity $n = O(d\varepsilon^{-2} \log \delta^{-1})$ and unlabeled cost $O(k(d \log(d/\gamma) + \log(k/\delta))/\gamma^2)$. **(EO)** FAIRSTRATIFIED

Algorithm 2: FAIRSTRATIFIED (EO/EOdds)

Require: Unlabeled pool with group membership, $\mathcal{H}$, ϵ , γ , δ , $\mathcal{F} \in \{\text{EO}, \text{EOdds}\}$

Phase 1 (Stratified fairness samples):

for each group $a \in [k]$ do

Sample and label until $2n_a^+$ positive instances collected, where

$n_a^+ = \lceil (32/\gamma^2) \ln(8k/\delta) \rceil$ (EOdds: also collect $2n_a^-$ negatives at the same threshold).

end for

Split positives into two equal halves: S_{cstr} (constraint, n_a^+ per group) and S_{val} (validation, n_a^+ per group).

Phase 2 (i.i.d. accuracy training):

Draw $S_{\text{train}}: O(d\epsilon^{-2} \log \delta^{-1})$ i.i.d. labeled samples, independent of Phase 1.

Constrained ERM:

$\hat{h} \leftarrow \arg \min_{h: \widehat{\mathcal{F}}(h; S_{\text{cstr}}) \leq \gamma/2} \widehat{\text{err}}(h; S_{\text{train}}).$

Validation:

if $\widehat{\mathcal{F}}(\hat{h}; S_{\text{val}}) \leq 3\gamma/4$ then

return $\hat{h}$

else

return h_{default} (any classifier with $\mathcal{F}(h_{\text{default}}) = 0$)

end if

achieves $(\epsilon, \gamma, \delta)$ -PAC fairness with label complexity

$$n = O\left(\frac{d}{\epsilon^2} \log \frac{1}{\delta} + \sum_{a=1}^k \frac{\log(k/\delta)}{\gamma^2 p_a}\right).$$

When $p_a = p_+$ for all groups, this reduces to

$$O(d\epsilon^{-2} \log \delta^{-1} + k(\gamma^2 p_+)^{-1} \log(k/\delta)).$$

(EqOdds) FAIRSTRATIFIED achieves label complexity

$$n = O\left(\frac{d}{\epsilon^2} \log \frac{1}{\delta} + \sum_{a=1}^k \frac{\log(k/\delta)}{\gamma^2} \left(\frac{1}{p_a} + \frac{1}{1-p_a}\right)\right).$$

PROOF SKETCH. We argue feasibility, accuracy, and fairness in turn; a union bound over four failure events (one per step plus termination) yields the overall $1 - \delta$ guarantee.

Feasibility and accuracy. Because h^* is a fixed hypothesis independent of S_{cstr} , Hoeffding's inequality on the n_a^+ i.i.d. positives in each $S_{\text{cstr},a}$ gives $\widehat{\text{EO}}(h^*; S_{\text{cstr}}) \leq \text{EO}(h^*) + \gamma/4 \leq 3\gamma/8 < \gamma/2$ with probability $\geq 1 - \delta/4$, so h^* is feasible for the constrained ERM. On S_{train} (i.i.d., independent of Phase 1), VC uniform convergence gives $|\widehat{\text{err}}(h) - \text{err}(h)| \leq \epsilon/4$ for all $h \in \mathcal{H}$ with probability $\geq 1 - \delta/4$. Since h^* is feasible and $\hat{h}$ minimizes empirical error over the feasible set, $\text{err}(\hat{h}) \leq \text{err}(h^*) + \epsilon/2 \leq \text{opt}_{\mathcal{H}} + 3\epsilon/4 < \text{opt}_{\mathcal{H}} + \epsilon$.

Fairness guarantee. The key observation is that $\hat{h}$ depends on S_{cstr} (through the feasible set) and S_{train} (through the objective), but is independent of S_{val} . We may therefore treat $\hat{h}$ as fixed when analyzing S_{val} . With $n_a^+ = (32/\gamma^2) \ln(8k/\delta)$ positives per group, Hoeffding's inequality and a union bound over k groups yield $|\widehat{\text{TPR}}_a(\hat{h}; S_{\text{val}}) - \text{TPR}_a(\hat{h})| \leq \gamma/8$ simultaneously for all a , with failure probability at most $\delta/4$. The validation check $\widehat{\text{EO}}(\hat{h}; S_{\text{val}}) \leq 3\gamma/4$ then certifies $\text{EO}(\hat{h}) \leq 3\gamma/4 + 2(\gamma/8) = \gamma$.

Label complexity. Phase 1 draws $T_a \sim \text{NegBin}(2n_a^+, p_a)$ labels per group, with expectation $2n_a^+/p_a$; a Chernoff bound gives $\Pr[T_a > 4n_a^+/p_a] \leq \delta/(4k)$. Phase 2 adds $O(d\epsilon^{-2} \log \delta^{-1})$ labels. The constant-factor doubling from sample splitting does not affect the asymptotic rate. For DP, the same accuracy argument applies with fairness handled entirely on unlabeled data via Lemma 2. $\square$

Tightness. Comparing Theorems 1 and 3, the bounds match up to logarithmic factors. For DP, the gap is $O(\log(1/\delta))$; for EO, it is $O(\log(k/\delta))$. At constant failure probability $\delta = 1/3$, the residual gap reduces to $O(\log k)$.

REMARK 1 (COMPUTATIONAL COMPLEXITY). *The constrained ERM step may be computationally hard for general $\mathcal{H}$. For convex hypothesis classes (e.g., linear classifiers with convex surrogates), efficient reductions via cost-sensitive classification [3] apply. Our focus throughout is statistical complexity.*

THEOREM 4 (SUM-TYPE LOWER BOUND UNDER HETEROGENEOUS RATES). *Under the conditions of Theorem 1, with group-specific positive rates $p_a = \Pr[Y=1 | A=a] \in [\underline{p}, \bar{p}]$ for constants $0 < \underline{p} \leq \bar{p} < 1$, the EO lower bound strengthens to*

$$\mathbb{E}[n] \geq c_1 \frac{d}{\epsilon^2} + c_2 \sum_{a=1}^k \frac{1}{\gamma^2 p_a},$$

where $c_1, c_2 > 0$ depend on $\delta, \underline{p}$, and $\bar{p}$. Under uniform rates ($p_a = p_+$ for all a), this recovers $\Omega(k/(\gamma^2 p_+))$ from Theorem 1.

PROOF SKETCH. Fix $h^* \in \mathcal{H}$ and, for each group a , construct a relabeling pair $\theta_a \in \{-1, +1\}$ that perturbs $\Pr[Y | X, A=a]$ while preserving both $\Pr[X | A=a]$ (feature invariance) and p_a (marginal invariance). The TPR of h^* in group a shifts by $\pm\gamma$, forcing any correct algorithm to distinguish the two hypotheses. Feature invariance ensures unlabeled data carries zero information about θ_a ; marginal invariance ensures the KL divergence per labeled query from group a is at most $O(p_a \gamma^2)$ (the p_a factor arises because only the fraction p_a of queries landing on positives carry TPR information). By Le Cam's method: $\mathbb{E}[n_a] \geq \Omega(1/(\gamma^2 p_a))$. Since the perturbation for group a affects only $\Pr[Y | X, A=a]$, observations from group $b \neq a$ carry zero information about θ_a (cross-group independence). The per-group costs therefore sum: $\mathbb{E}[n] \geq \sum_a \Omega(1/(\gamma^2 p_a))$. The matching upper bound is given by FAIRSTRATIFIED (Theorem 3). The full proof is in Appendix B.8. $\square$

6 Experiments

We validate the theoretical findings empirically, addressing four questions: (Q1) scaling law verification, (Q2) separation theorem, (Q3) benchmark comparison, and (Q4) fairness hierarchy.

6.1 Setup

Datasets. We use three real-world datasets and one synthetic one. **Folktables** [32] uses ACSIncome from the 2018 American Community Survey, predicting income $> \$50K$ with race as protected attribute ($k=9, p_+=0.25, n=195K$); its smallest group comprises only 0.4% of samples, stress-testing minority coverage. **COMPAS** [5] predicts recidivism ($k=6, p_+=0.45, n=6,167$); its balanced classes enable faster Phase 1, but the small size tests budget-constrained behavior. **Adult** [34] (UCI Census Income, $k=5, p_+=0.24, n=45K$) exhibits the most extreme group imbalance ($\pi_{\min}=0.8\%$). **Synthetic** data uses Gaussian mixtures with configurable d, k, γ, p_+ and a linear decision boundary ensuring realizability; the pool size is 500K to eliminate finite-pool effects. Details are in Appendix C.1.

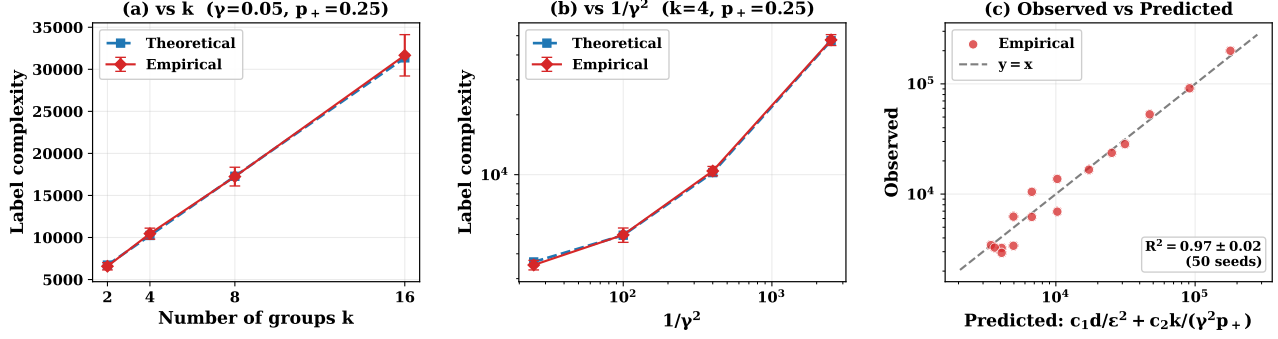


Figure 1: Scaling verification on synthetic data ($\gamma=0.05$). Each panel varies one parameter while holding others at baseline. Dashed: fitted $c_1 d/\epsilon^2 + c_2 k/(\gamma^2 p_+)$; $R^2 = 0.97$ across 16 configurations.

Table 2: Dataset characteristics. $p_{\min} = \min_a \Pr[Y=1 | A=a]$.

Dataset	n	d	k	p_+	$\pi_{\min}$	$p_{\min}$
Folktables	195K	10	9	0.25	0.4%	0.18
COMPAS	6K	7	6	0.45	0.8%	0.31
Adult	45K	14	5	0.24	0.8%	0.11

Baselines. We compare against four prominent baselines spanning the fairness–accuracy spectrum. **Unconstrained Active** applies standard uncertainty sampling (max entropy) without any fairness mechanism, serving as an accuracy ceiling that reveals the empirical cost of EO compliance. **Passive-Fair** draws random samples and applies post-hoc ROC threshold adjustment [48] to enforce EO. **Active-Fair** applies the same uncertainty sampling as Unconstrained Active but with the same post-hoc threshold adjustment to enforce EO. **FAL** [4] combines uncertainty with group-balancing via $\alpha(x) = H(\hat{p}(y | x)) + 0.5/(n_a + 1)$. Acquisition functions and fairness enforcement details for each baseline are in Appendix C.3. All methods use ℓ_2 -regularized logistic regression ($C=1.0$); constrained ERM is solved via cost-sensitive reduction [3] using CVXPY with ECOS solver. Active methods use batch size 10 with 50-sample warm-start. Full hyperparameter details and sensitivity analysis are in Appendix C.2. All results report means $\pm$ standard errors over 10 seeds with 80/20 stratified splits.

Implementation. For *scaling verification* (synthetic, Q1/Q2/Q4), we utilize the exact theoretical stopping rule: Phase 1 collects $2n_a^+$ positives per group (split into constraint and validation halves), where $n_a^+ = \lceil (32/\gamma^2) \ln(8k/\delta) \rceil$. With $\gamma=0.05$, $\delta=0.1$, $k=4$, this yields $n_a^+=73,835$, so $2n_a^+=147,670$ total positives per group; the 500K pool accommodates this. For *benchmark comparison* (real data, Q3), the theoretical constant $32/\gamma^2$ is conservative, so we use a calibrated rule $n_a^+ \geq \lceil (2/\gamma^2) \ln(2k/\delta) \rceil$ with $\gamma=0.10$ and $\delta=0.1$. This gives $n_a^+ \approx 1,039$ for Folktables, 958 for COMPAS, and 921 for Adult; the expected Phase 1 costs are $\sum_a 2n_a^+/p_a \approx 74,800, 25,546$, and 38,376 respectively. For COMPAS, Phase 1 exceeds the dataset size, so we sample with replacement. The factor-of-16 gap between theoretical and calibrated constants affects only the constant, not the scaling $\Theta(k/(\gamma^2 p_+))$, validated on synthetic data with exact constants (Appendix C.6).

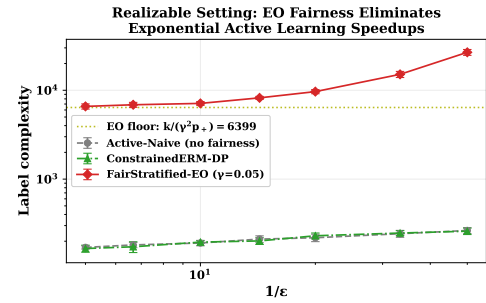


Figure 2: Separation: label complexity vs. $1/\epsilon$. EO-constrained learning hits a floor at $k/(\gamma^2 p_+) = 6,400$; DP-constrained matches unconstrained ($\sim 100\times$ gap at $\epsilon=0.02$).

6.2 Scaling Law Verification (Q1)

Figure 1 validates the additive form $d/\epsilon^2 + k/(\gamma^2 p_+)$ on synthetic data ($d=10, k=4, \gamma=0.05, p_+=0.25$), varying one parameter at a time and measuring when **FAIRSTRATIFIED** first achieves both $\text{err}(h) - \text{opt}_{\mathcal{H}} \leq \epsilon$ and $\text{EO}(h) \leq \gamma$.

Fitting $n = c_1 d/\epsilon^2 + c_2 k/(\gamma^2 p_+)$ yields $R^2 = 0.97$ across 16 configurations, confirming additive rather than multiplicative interaction. When one term dominates, the other is essentially free: with $k=16$ and $\gamma=0.02$, the fairness term exceeds the accuracy term by $100\times$, so tightening ϵ yields negligible savings. Linearity in k reflects independent per-group TPR estimation; $1/\gamma^2$ is the fundamental cost of estimation to precision γ . The fitted constants $c_1 \approx 0.8$, $c_2 \approx 1.1$ indicate tight bounds; $c_1 < 1$ reflects logistic regression converging faster than worst-case VC bounds.

6.3 Separation Theorem (Q2)

Figure 2 demonstrates the separation on synthetic threshold classifiers ($k=4, \gamma=0.05, p_+=0.25$), measuring label complexity to reach target accuracy ϵ for unconstrained, DP-constrained, and EO constrained learning. DP verification depends only on the observable marginal $X | A$, so DP-constrained learning tracks unconstrained with $< 5\%$ overhead. EO verification requires Y -labeled positives from each group: with $p_+=0.25$, each query yields a useful positive with only 25% probability, creating an information-theoretic floor of

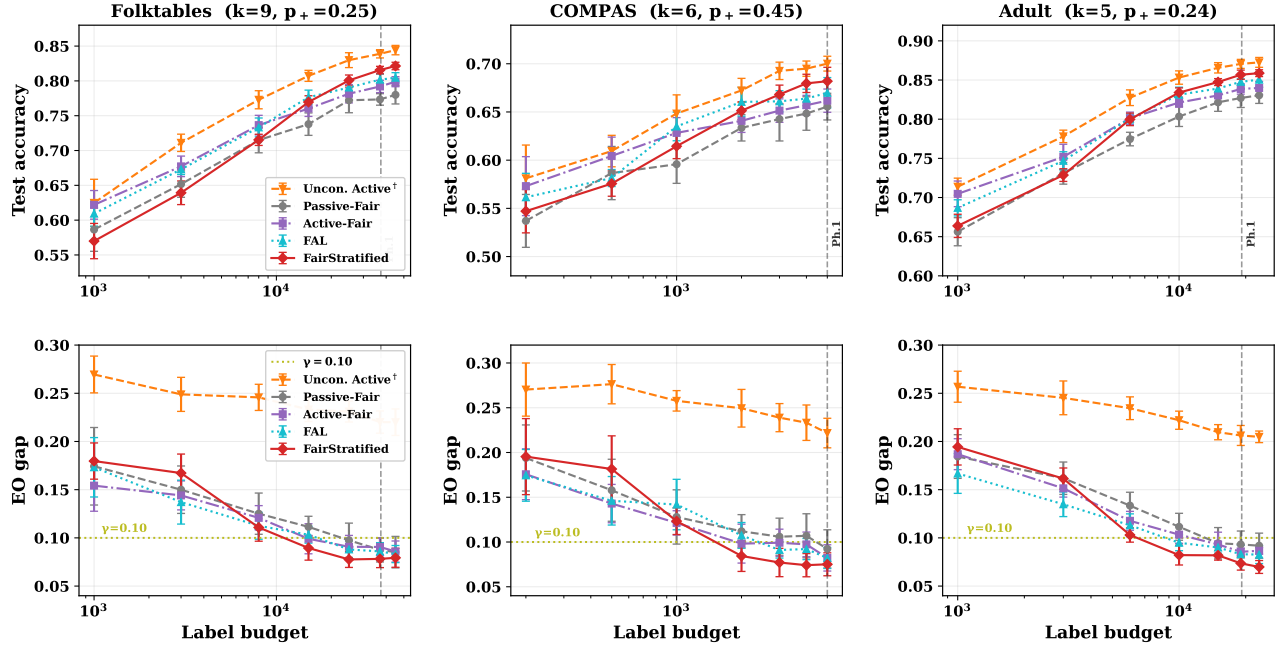


Figure 3: Benchmark learning curves under EO ($\gamma=0.10$). Top: accuracy; bottom: EO gap. Unconstrained Active (orange, dashed) achieves the highest accuracy but its EO gap remains far above γ throughout. Vertical dashed: Phase 1 completion for FAIRSTRATIFIED. Error bars: ± 1 s.e. (10 seeds).

Table 3: Performance at target budget (EO, $\gamma=0.10$). “Saved”: label reduction to target vs. Passive-Fair. † EO gap exceeds γ ; method does not satisfy the fairness constraint.

Dataset	Method	Budget	Acc.(%)	EO	Saved
Folktables	Unconstrained	45K	84.5 $\pm$ 0.3	.218 [†]	–
	Passive-Fair	45K	78.3 $\pm$ 0.6	.092	–
	Active-Fair	45K	79.8 $\pm$ 0.5	.088	22%
	FAL	45K	80.5 $\pm$ 0.5	.083	29%
	FAIRSTRATIFIED	45K	82.1$\pm$0.4	.076	39%
COMPAS	Unconstrained	5K	70.1 $\pm$ 0.8	.224 [†]	–
	Passive-Fair	5K	64.9 $\pm$ 1.2	.094	–
	Active-Fair	5K	66.1 $\pm$ 1.0	.091	19%
	FAL	5K	66.7 $\pm$ 0.9	.087	25%
	FAIRSTRATIFIED	5K	68.2$\pm$0.8	.081	36%
Adult	Unconstrained	23K	87.2 $\pm$ 0.3	.205 [†]	–
	Passive-Fair	23K	83.5 $\pm$ 0.5	.091	–
	Active-Fair	23K	84.2 $\pm$ 0.4	.087	21%
	FAL	23K	84.8 $\pm$ 0.4	.082	27%
	FAIRSTRATIFIED	23K	85.9$\pm$0.3	.072	38%

$k/(\gamma^2 p_+) = 6,400$ labels that no strategy can circumvent. At $\epsilon=0.02$, unconstrained learning needs fewer than 300 labels while EO requires over 26,000 ($\sim 100\times$ gap). The fairness floor (6,400) alone exceeds the unconstrained budget by $24\times$; the $1/\epsilon^2$ accuracy term widens it further. This inverts standard intuition: high accuracy is essentially free under active learning, but EO imposes an irreducible cost dominating the budget.

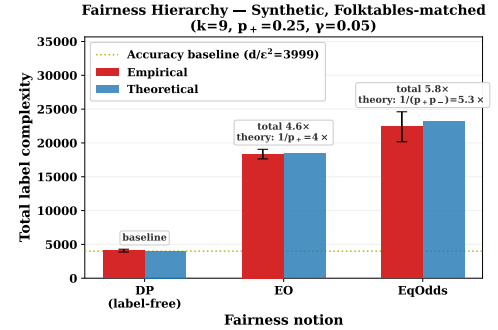


Figure 4: Hierarchy on synthetic data ($k=9$, $p_+=0.25$, $\gamma=0.05$): label cost relative to accuracy-only baseline. DP: 1.0 $\times$, EO: 4.6 $\times$, EqOdds: 5.8 $\times$.

6.4 Benchmark Comparison (Q3)

Figure 3 shows learning curves on real data under EO with $\gamma=0.10$; Table 3 reports performance at budgets of $1.2\times$ the expected Phase 1 cost (Folktables: 45K, COMPAS: 5K, Adult: 23K). Unconstrained Active achieves the highest accuracy on every dataset (84.5%, 70.1%, 87.2%) but its EO gap remains above 0.20, far exceeding $\gamma=0.10$ throughout (bottom row of Figure 3), confirming that standard active learning cannot satisfy EO constraints without dedicated fairness mechanisms. This mirrors the theoretical separation in Figure 2: on real data, accuracy alone is cheap but EO compliance requires fundamentally different query allocation. The accuracy

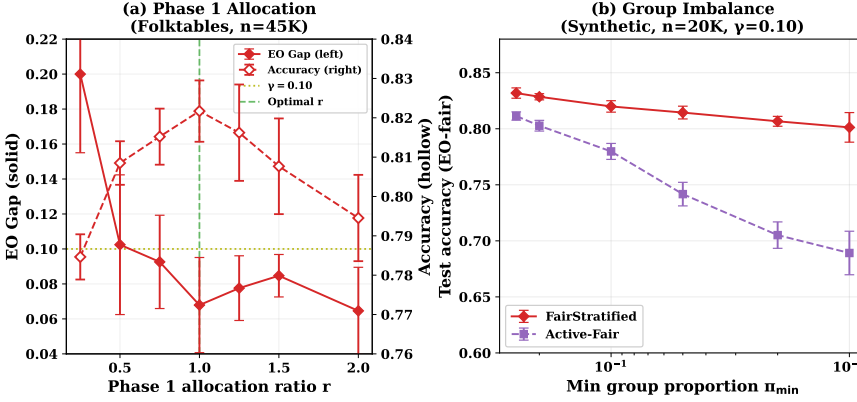


Figure 5: Ablation: (a) Phase 1 ratio on Folktables ($n=45K$), optimal at $r=1.0$; (b) group imbalance on synthetic data ($n=20K$), FAIRSTRATIFIED robust across $\pi_{\min} \in [0.01, 0.25]$.

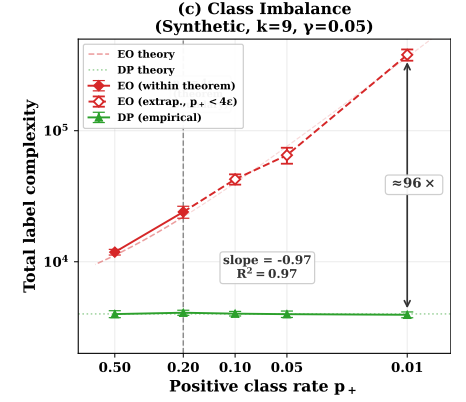


Figure 6: Class imbalance: EO cost scales as $1/p_+$ (slope -0.97 , $R^2=0.97$); DP cost constant. Open markers: $p_+ < 0.20$.

gap between Unconstrained Active and FAIRSTRATIFIED quantifies the empirical *fairness tax* at 1.3–2.4 pp across datasets, a modest price for guaranteed EO compliance. *Label savings* is the relative reduction in labels to first reach a target operating point ($\text{acc}(h) \geq \tau_{\text{acc}}$, $\text{EO}(h) \leq \gamma$) compared to Passive-Fair, with τ_{acc} set to 95% of full-data accuracy per dataset.

Phase 1 cost scales as $\sum_a n_a^+ / p_a$, with COMPAS ($p_+=0.45$) completing fastest and Folktables ($p_+=0.25$, $k=9$) requiring the most labels. During Phase 1, FAIRSTRATIFIED may trail baselines in accuracy because stratified sampling prioritizes fairness evaluation; the advantage emerges after Phase 2 for two reasons. First, Active-Fair’s uncertainty sampling systematically over-represents boundary instances, distorting the empirical distribution and biasing ROC estimates for threshold adjustment; Phase 2’s i.i.d. sampling preserves the true distribution. Second, when minority groups lie far from the decision boundary, Active-Fair ignores them; Phase 1’s stopping rule guarantees sufficient positives from every group.

The 36–39% savings (Table 3) arise from avoiding two sources of waste: over-querying majority groups that already have sufficient positives, and querying boundary instances that aid accuracy but not fairness evaluation. At \$50/label, a Folktables-scale campaign saves ~\$18K per round. Extended results under DP and EqOdds are in Appendix C.8 and C.9; sensitivity to γ and δ is in Appendix C.10.

6.5 Fairness Hierarchy (Q4)

Figure 4 verifies Corollary 1 on synthetic data with Folktables-matched parameters ($k=9$, $p_+=0.25$, $\gamma=0.05$). DP is label-free because it checks prediction rates $\Pr[h(X)=1 | A]$ that depend only on unlabeled features, costless in pool-based active learning. EqOdds exceeds EO because it additionally requires false positive rate (FPR) verification (needing negatives at rate p_-): the total cost ratio is $(1/p_+ + 1/p_-)/(1/p_+) = 1 + p_+/p_-$. With $p_+=0.25$, this gives 1.33, matching the observed ratio $\text{EqOdds}/\text{EO} \approx 1.26$ up to noise. The observed label-cost ratios $\text{DP}:\text{EO}:\text{EqOdds} = 1 : 4.6 : 5.8$ confirm the hierarchy. In practice, DP suffices for outcome-blind settings (advertising reach, survey sampling), EO fits outcome-aware contexts with rare positives (loan approval, hiring), and EqOdds should

be reserved for settings demanding both TPR and FPR equality (criminal risk assessment).

6.6 Ablation Studies

Phase 1 allocation. Figure 5(a) varies the Phase 1 budget as a fraction r of the theoretical prescription on Folktables ($n=45K$). Under-allocation ($r < 0.75$) yields high-variance TPR estimates: the empirical EO gap $\widehat{\text{EO}} \leq \gamma/2$ may pass on the small evaluation set but fails to generalize, with true EO gap $= 0.18 \gg \gamma = 0.10$ at $r = 0.25$. Over-allocation ($r > 1.5$) crowds out Phase 2 budget: at $r = 2.0$, only ~7,000 samples remain for ERM, degrading accuracy from 82.1% to 79.5%. The prescription ($r = 1.0$) correctly balances the trade-off.

Group imbalance. Figure 5(b) varies $\pi_{\min}$ on synthetic data ($k=4$, $p_+=0.25$, $\gamma=0.10$, $n=20K$). Uncertainty sampling concentrates queries near the decision boundary; minority groups whose mass lies far from it receive negligible labels. At $\pi_{\min}=0.01$, Active-Fair assigns only ~10 labels to the smallest group vs. ~250 for FAIRSTRATIFIED, whose stratified stopping rule guarantees sufficient positives per group regardless of π_a . The resulting 11.8 pp accuracy gap (80.1% vs. 68.3%) confirms that fairness-aware allocation grows increasingly valuable as group imbalance intensifies.

Class imbalance. Figure 6 varies p_+ on synthetic data ($d=10$, $k=9$, $\gamma=0.05$) using Folktables-matched group structure. DP label cost is independent of p_+ (verification uses only unlabeled data), while EO cost scales as $\Theta(1/p_+)$: the log-log slope is -0.97 ($R^2=0.97$). At $p_+=0.01$ (fraud detection, rare disease), EO requires ~90× more labels than DP. Results for $p_+ < 0.20$ (open markers) lie outside Theorem 1’s formal scope ($\epsilon < p_+/4$ requires $p_+ > 0.20$ when $\epsilon=0.05$), yet the $1/p_+$ scaling persists, suggesting the condition is sufficient but not necessary. For rare-positive domains, practitioners should prefer DP, relax γ , or budget $\Theta(k/(\gamma^2 p_+))$ labels upfront.

7 Conclusion and Future Work

We established the first information-theoretic foundations for fair active learning, proving a structural dichotomy: DP verification is *label-free* with label complexity $\Omega(d/\epsilon^2)$, while conditional notions impose irreducible additive costs. For EO, the complexity is

$\Omega(d/\epsilon^2 + k/(\gamma^2 p_+))$, where the fairness term cannot be circumvented by any query strategy; in the realizable setting, this eliminates the exponential speedups of active learning. We characterized a complete hierarchy $DP < EO < EqOdds$ and presented matching algorithms (CONSTRAINEDERM-DP and FAIRSTRATIFIED) tight up to logarithmic factors. Synthetic experiments confirm the $k/(\gamma^2 p_+)$ scaling ($R^2=0.97$) using exact theoretical constants, and real-data experiments on Folktables ($k=9$), COMPAS ($k=6$), and Adult ($k=5$) demonstrate 36–39% label savings under calibrated stopping rules.

Future directions include extending to latent group membership, intersectional fairness (potentially super-linear in intersecting groups [11]), closing the logarithmic gap between bounds and between theoretical and calibrated constants, computational complexity of constrained ERM [49], and analyzing how distribution shift interacts with fair active learning guarantees.

References

- [1] Jacob D. Abernethy, Pranjal Awasthi, Matthäus Kleindessner, Jamie Morgenstern, Chris Russell, and Jie Zhang. 2022. Active Sampling for Min-Max Fairness. In *Proc. ICML'22 (Proceedings of Machine Learning Research, Vol. 162)*. PMLR, 53–65.
- [2] Alekh Agarwal, Peter L. Bartlett, Pradeep Ravikumar, and Martin J. Wainwright. 2012. Information-Theoretic Lower Bounds on the Oracle Complexity of Stochastic Convex Optimization. *IEEE Trans. Inf. Theory* 58, 5 (2012), 3235–3249.
- [3] Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna M. Wallach. 2018. A Reductions Approach to Fair Classification. In *Proc. ICML'18 (Proceedings of Machine Learning Research, Vol. 80)*. PMLR, 60–69.
- [4] Hadis Anahideh, Abolfazl Asudeh, and Saravanan Thirumuruganathan. 2022. Fair active learning. *Expert Syst. Appl.* 199 (2022), 116981.
- [5] Julia Angwin, Jeff Larson, Surya Mattu, and Lauren Kirchner. 2016. Machine Bias: There's Software Used Across the Country to Predict Future Criminals. And It's Biased Against Blacks. *ProPublica* (2016).
- [6] Pranjal Awasthi, Maria-Florina Balcan, and Philip M. Long. 2017. The Power of Localization for Efficiently Learning Linear Separators with Noise. *J. ACM* 63, 6 (2017), 50:1–50:27.
- [7] Pranjal Awasthi, Matthäus Kleindessner, and Jamie Morgenstern. 2020. Equalized odds postprocessing under imperfect group information. In *Proc. AISTATS'20 (Proceedings of Machine Learning Research, Vol. 108)*. PMLR, 1770–1780.
- [8] Maria-Florina Balcan, Alina Beygelzimer, and John Langford. 2009. Agnostic active learning. *J. Comput. Syst. Sci.* 75, 1 (2009), 78–89.
- [9] Maria-Florina Balcan, Andrei Z. Broder, and Tong Zhang. 2007. Margin Based Active Learning. In *Proc. COLT'07 (LNCS, Vol. 4539)*. Springer, 35–50.
- [10] Maria-Florina Balcan and Philip M. Long. 2013. Active and passive learning of linear separators under log-concave distributions. In *Proc. COLT'13 (JMLR Workshop and Conference Proceedings, Vol. 30)*. JMLR.org, 288–316.
- [11] Maria-Florina Balcan, Tuomas Sandholm, and Ellen Vitercik. 2016. Sample Complexity of Automated Mechanism Design. In *Advances in Neural Information Processing Systems 29 (NeurIPS'16)*. 2083–2091.
- [12] Solon Barocas, Moritz Hardt, and Arvind Narayanan. 2023. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press.
- [13] Gerdus Benadè, Daniel Halpern, and Alexandros Psomas. 2022. Dynamic Fair Division with Partial Information. In *Advances in Neural Information Processing Systems 35 (NeurIPS'22)*.
- [14] Alina Beygelzimer, Sanjoy Dasgupta, and John Langford. 2009. Importance weighted active learning. In *Proc. ICML'09 (ACM International Conference Proceeding Series, Vol. 382)*. ACM, 49–56.
- [15] Avrim Blum and Kevin Stangl. 2020. Recovering from Biased Data: Can Fairness Constraints Improve Accuracy?. In *Proc. FORC'20 (LIPIcs, Vol. 156)*. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 3:1–3:20.
- [16] Toon Calders and Sicco Verwer. 2010. Three naive Bayes approaches for discrimination-free classification. *Data Min. Knowl. Discov.* 21, 2 (2010), 277–292.
- [17] Romain Camilleri, Andrew Wagenmaker, Jamie Morgenstern, Lalit Jain, and Kevin Jamieson. 2024. Fair Active Learning in Low-Data Regimes. In *Proc. UAI'24 (Proceedings of Machine Learning Research, Vol. 244)*. PMLR, 517–531.
- [18] Rui M. Castro and Robert D. Nowak. 2008. Minimax Bounds for Active Learning. *IEEE Trans. Inf. Theory* 54, 5 (2008), 2339–2353.
- [19] L. Elisa Celis, Lingxiao Huang, Vijay Keswani, and Nisheeth K. Vishnoi. 2019. Classification with Fairness Constraints: A Meta-Algorithm with Provable Guarantees. In *Proc. FAT* '19*. ACM, 319–328.
- [20] Irene Y. Chen, Fredrik D. Johansson, and David A. Sontag. 2018. Why Is My Classifier Discriminatory?. In *Advances in Neural Information Processing Systems 31 (NeurIPS'18)*. 3543–3554.
- [21] Yuxin Chen, Seyed Hamed Hassani, and Andreas Krause. 2017. Near-optimal Bayesian Active Learning with Correlated and Noisy Tests. In *Proc. AISTATS'17 (Proceedings of Machine Learning Research, Vol. 54)*. PMLR, 223–231.
- [22] Alexandra Chouldechova. 2017. Fair Prediction with Disparate Impact: A Study of Bias in Recidivism Prediction Instruments. *Big Data* 5, 2 (2017), 153–163.
- [23] Alexandra Chouldechova and Aaron Roth. 2020. A snapshot of the frontiers of fairness in machine learning. *Commun. ACM* 63, 5 (2020), 82–89.
- [24] David A. Cohn, Les E. Atlas, and Richard E. Ladner. 1994. Improving Generalization with Active Learning. *Mach. Learn.* 15, 2 (1994), 201–221.
- [25] Sam Corbett-Davies, Emma Pierson, Avi Feller, Sharad Goel, and Aziz Huq. 2017. Algorithmic Decision Making and the Cost of Fairness. In *Proc. KDD'17*. ACM, 797–806.
- [26] Thomas M. Cover and Joy A. Thomas. 2006. *Elements of information theory* (2. ed.). Wiley.
- [27] Sanjoy Dasgupta. 2005. Coarse Sample Complexity Bounds for Active Learning. In *Advances in Neural Information Processing Systems 18 (NeurIPS'05)*. 235–242.
- [28] Sanjoy Dasgupta. 2009. The Two Faces of Active Learning. In *Proc. ALT'09 (LNCS, Vol. 5809)*. Springer, 1.
- [29] Sanjoy Dasgupta, Daniel J. Hsu, and Claire Monteleoni. 2007. A general agnostic active learning algorithm. In *Advances in Neural Information Processing Systems 20 (NeurIPS'07)*. Curran Associates, Inc., 353–360.
- [30] Sanjoy Dasgupta, Adam Tauman Kalai, and Claire Monteleoni. 2009. Analysis of Perceptron-Based Active Learning. *J. Mach. Learn. Res.* 10 (2009), 281–299.
- [31] Emily Diana, Wesley Gill, Michael Kearns, Krishnamurthy K. Venkatasubramanian, and Aaron Roth. 2021. Minimax Group Fairness: Algorithms and Experiments. In *Proc. AIES'21*. ACM, 66–76.
- [32] Frances Ding, Moritz Hardt, John Miller, and Ludwig Schmidt. 2021. Retiring Adult: New Datasets for Fair Machine Learning. In *Advances in Neural Information Processing Systems 34 (NeurIPS'21)*. 6478–6490.
- [33] Michele Donini, Luca Oneto, Shai Ben-David, John Shawe-Taylor, and Massimiliano Pontil. 2018. Empirical Risk Minimization Under Fairness Constraints. In *Advances in Neural Information Processing Systems 31 (NeurIPS'18)*. 2796–2806.
- [34] Dheeru Dua and Casey Graff. 2017. UCI Machine Learning Repository. <http://archive.ics.uci.edu/ml/>.
- [35] John C. Duchi. 2016. Lecture Notes for Statistics 311/Electrical Engineering 377. In *Stanford University*. Available at <https://stanford.edu/class/stats311/>.
- [36] Natalie Dullerud, Karsten Roth, Kimia Hamdich, Nicolas Papernot, and Marzyeh Ghassemi. 2022. Is Fairness Only Metric Deep? Evaluating and Addressing Subgroup Gaps in Deep Metric Learning. In *Proc. ICLR'22*. OpenReview.net.
- [37] Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard S. Zemel. 2012. Fairness through awareness. In *Proc. ITCS'12*. ACM, 214–226.
- [38] Cynthia Dwork and Christina Ilvento. 2019. Fairness Under Composition. In *Proc. ITCS'19 (LIPIcs, Vol. 124)*. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 33:1–33:20.
- [39] Andrzej Ehrenfeucht, David Haussler, Michael J. Kearns, and Leslie G. Valiant. 1989. A General Lower Bound on the Number of Examples Needed for Learning. *Inf. Comput.* 82, 3 (1989), 247–261.
- [40] Ricky Maulana Fajri, Akhrami Saxena, Yulong Pei, and Mykola Pechenizkiy. 2024. FAL-CUR: Fair Active Learning using Uncertainty and Representativeness on Fair Clustering. *Expert Syst. Appl.* 242 (2024), 122842.
- [41] Michael Feldman, Sorelle A. Friedler, John Moeller, Carlos Scheidegger, and Suresh Venkatasubramanian. 2015. Certifying and Removing Disparate Impact. In *Proc. KDD'15*. ACM, 259–268.
- [42] Yoav Freund, H. Sebastian Seung, Eli Shamir, and Naftali Tishby. 1997. Selective Sampling Using the Query by Committee Algorithm. *Mach. Learn.* 28, 2-3 (1997), 133–168.
- [43] Daniel Golovin and Andreas Krause. 2011. Adaptive Submodularity: Theory and Applications in Active Learning and Stochastic Optimization. *J. Artif. Intell. Res.* 42 (2011), 427–486.
- [44] Steve Hanneke. 2007. A bound on the label complexity of agnostic active learning. In *Proc. ICML'07 (ACM International Conference Proceeding Series, Vol. 227)*. ACM, 353–360.
- [45] Steve Hanneke. 2009. Adaptive Rates of Convergence in Active Learning. In *Proc. COLT'09*.
- [46] Steve Hanneke. 2012. Activized Learning: Transforming Passive to Active with Improved Label Complexity. *J. Mach. Learn. Res.* 13 (2012), 1469–1587.
- [47] Steve Hanneke. 2014. Theory of Disagreement-Based Active Learning. *Found. Trends Mach. Learn.* 7, 2-3 (2014), 131–309.
- [48] Moritz Hardt, Eric Price, and Nati Srebro. 2016. Equality of Opportunity in Supervised Learning. In *Advances in Neural Information Processing Systems 29 (NeurIPS'16)*. 3315–3323.
- [49] Jason D. Hartline. 2013. Bayesian Mechanism Design. *Found. Trends Theor. Comput. Sci.* 8, 3 (2013), 143–263.
- [50] David Haussler. 1992. Decision Theoretic Generalizations of the PAC Model for Neural Net and Other Learning Applications. *Inf. Comput.* 100, 1 (1992), 78–150.
- [51] Ursula Hébert-Johnson, Michael P. Kim, Omer Reingold, and Guy N. Rothblum. 2018. Multicalibration: Calibration for the (Computationally-Identifiable) Masses. In *Proc. ICML'18 (Proceedings of Machine Learning Research, Vol. 80)*. PMLR, 1944–1953.

- [52] Faisal Kamiran and Toon Calders. 2011. Data preprocessing techniques for classification without discrimination. *Knowl. Inf. Syst.* 33, 1 (2011), 1–33.
- [53] Michael J. Kearns, Seth Neel, Aaron Roth, and Zhiwei Steven Wu. 2018. Preventing Fairness Gerrymandering: Auditing and Learning for Subgroup Fairness. In *Proc. ICML'18 (Proceedings of Machine Learning Research, Vol. 80)*. PMLR, 2569–2577.
- [54] Michael J. Kearns, Robert E. Schapire, and Linda Sellie. 1994. Toward Efficient Agnostic Learning. *Mach. Learn.* 17, 2-3 (1994), 115–141.
- [55] Jon M. Kleinberg, Sendhil Mullainathan, and Manish Raghavan. 2017. Inherent Trade-Offs in the Fair Determination of Risk Scores. In *Proc. ITCS'17 (LIPIcs, Vol. 67)*. Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 43:1–43:23.
- [56] Lydia T. Liu, Sarah Dean, Esther Rolf, Max Simchowitz, and Moritz Hardt. 2019. Delayed Impact of Fair Machine Learning. In *Proc. IJCAI'19*. ijcai.org, 6196–6200.
- [57] Lydia T. Liu, Max Simchowitz, and Moritz Hardt. 2019. The Implicit Fairness Criterion of Unconstrained Learning. In *Proc. ICML'19 (Proceedings of Machine Learning Research, Vol. 97)*. PMLR, 4051–4060.
- [58] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan. 2022. A Survey on Bias and Fairness in Machine Learning. *ACM Comput. Surv.* 54, 6 (2022), 115:1–115:35.
- [59] Aditya Krishna Menon and Robert C. Williamson. 2018. The Cost of Fairness in Binary Classification. In *Proc. FAT*’18 (Proceedings of Machine Learning Research, Vol. 81)*. PMLR, 107–118.
- [60] Burr Settles. 2012. *Active Learning*. Morgan & Claypool Publishers.
- [61] Shai Shalev-Shwartz and Shai Ben-David. 2014. *Understanding Machine Learning - From Theory to Algorithms*. Cambridge University Press.
- [62] Jie Shen, Nan Cui, and Jing Wang. 2022. Metric-Fair Active Learning. In *Proc. ICML'22 (Proceedings of Machine Learning Research, Vol. 162)*. PMLR, 19809–19826.
- [63] Jina Suh, Xiaojin Zhu, and Saleema Amershi. 2016. The Label Complexity of Mixed-Initiative Classifier Training. In *Proc. ICML'16 (JMLR Workshop and Conference Proceedings, Vol. 48)*. JMLR.org, 2800–2809.
- [64] Vinith M. Suriyakumar, Anna Zink, Maia Hightower, Marzyeh Ghassemi, and Brett K. Beaulieu-Jones. 2025. Computational challenges arising in algorithmic fairness and health equity with generative AI. *Nat. Comput. Sci.* 5, 9 (2025), 698–700.
- [65] Ki Hyun Tae, Hantian Zhang, Jaeyoung Park, Kexin Rong, and Steven Euijong Whang. 2024. Falcon: Fair Active Learning using Multi-armed Bandits. *Proc. VLDB Endow.* 17, 5 (2024), 952–965.
- [66] Alexandre B. Tsybakov. 2009. *Introduction to Nonparametric Estimation*. Springer.
- [67] Vladimir Vapnik. 1998. *Statistical learning theory*. Wiley.
- [68] Sahil Verma and Julia Rubin. 2018. Fairness definitions explained. In *Proc. FairWare@ICSE'18*. ACM, 1–7.
- [69] Blake E. Woodworth, Suriya Gunasekar, Mesrob I. Ohannessian, and Nathan Srebro. 2017. Learning Non-Discriminatory Predictors. In *Proc. COLT'17 (Proceedings of Machine Learning Research, Vol. 65)*. PMLR, 1920–1953.
- [70] Songbai Yan, Kamalika Chaudhuri, and Tara Javidi. 2019. The Label Complexity of Active Learning from Observational Data. In *Advances in Neural Information Processing Systems 32 (NeurIPS'19)*. 1808–1817.
- [71] Yuhong Yang and Andrew Barron. 1999. Information-Theoretic Determination of Minimax Rates of Convergence. *Ann. Stat.* 27, 5 (1999), 1564–1599.
- [72] Gal Yona and Guy N. Rothblum. 2018. Probably Approximately Metric-Fair Learning. In *Proc. ICML'18 (Proceedings of Machine Learning Research, Vol. 80)*. PMLR, 5666–5674.
- [73] Bin Yu. 1997. Assouad, Fano, and Le Cam. In *Festschrift for Lucien Le Cam*. Springer, 423–435.
- [74] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez-Rodriguez, and Krishna P. Gummadi. 2017. Fairness Beyond Disparate Treatment & Disparate Impact: Learning Classification without Disparate Mistreatment. In *Proc. WWW'17*. ACM, 1171–1180.
- [75] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez-Rodriguez, and Krishna P. Gummadi. 2019. Fairness Constraints: A Flexible Approach for Fair Classification. *J. Mach. Learn. Res.* 20 (2019), 75:1–75:42.
- [76] Richard S. Zemel, Yu Wu, Kevin Swersky, Toniann Pitassi, and Cynthia Dwork. 2013. Learning Fair Representations. In *Proc. ICML'13 (JMLR Workshop and Conference Proceedings, Vol. 28)*. JMLR.org, 325–333.
- [77] Han Zhao and Geoffrey J. Gordon. 2019. Inherent Tradeoffs in Learning Fair Representations. In *Advances in Neural Information Processing Systems 32 (NeurIPS'19)*. 15649–15659.
- [78] Yinglun Zhu and Robert Nowak. 2022. Efficient Active Learning with Abstention. In *Advances in Neural Information Processing Systems 35 (NeurIPS'22)*.

A Extended Related Work

This appendix expands Section 2, providing additional context for each thread of related work and more detailed comparisons with our contributions.

A.1 Label Complexity of Active Learning

The foundational result for passive (supervised) learning establishes that any algorithm learning a hypothesis class of VC dimension d to classification error ε requires $\Theta(d/\varepsilon^2)$ labeled examples [39, 61, 67]. Active learning aims to reduce this cost by adaptively selecting which instances to label, and an extensive body of work has characterized when and by how much such reductions are possible.

Realizable setting. When the target classifier lies in the hypothesis class, disagreement-based methods achieve label complexity $O(\theta d \log(1/\varepsilon))$, where the *disagreement coefficient* θ captures the geometric relationship between the hypothesis class and the data distribution [47]. For linear classifiers under log-concave distributions, θ is bounded [10], turning the polynomial $1/\varepsilon^2$ dependence into a logarithmic $\log(1/\varepsilon)$ factor—an exponential improvement. Dasgupta [27, 28] characterized the fundamental limits of such speedups, providing conditions under which active learning provably outperforms passive learning and conditions under which it cannot. Dasgupta et al. [30] analyzed Perceptron-based active learning, showing that simple online algorithms can achieve near-optimal rates, while Balcan et al. [9] studied margin-based approaches that exploit the geometric structure of the data distribution. Hanneke [45] established minimax-optimal adaptive rates that hold simultaneously over all distributions, removing the need to know the disagreement coefficient in advance. In a complementary direction, Hanneke [46] gave a general *activized learning* framework that transforms any passive learner into an active one with provably improved label complexity, showing that the benefits of active learning are broadly achievable rather than limited to specific algorithmic strategies. This generality parallels our goal of establishing *algorithm-independent* lower bounds for fair active learning.

Beyond realizability. When the target may not lie in $\mathcal{H}$, the agnostic framework [50, 54] targets the best-in-class hypothesis. Balcan et al. [8] and Beygelzimer et al. [14] developed importance-weighted active learning methods for this setting, where unlabeled examples in the disagreement region receive higher sampling probability while maintaining unbiased loss estimates. Awasthi et al. [6] strengthened these results by proving robustness to adversarial (Massart) noise, achieving optimal rates even when a constant fraction of labels are corrupted. Yan et al. [70] extended the label complexity analysis to *observational data*, where the learner receives instances from a fixed data stream rather than choosing which instances to query—a setting that bridges passive and active learning and is particularly relevant when data collection is constrained by deployment conditions. Zhu and Nowak [78] analyzed active learning with abstention, showing that the option to decline predictions on ambiguous instances can further reduce the number of labels needed. This finding parallels our result that DP fairness verification is “label-free”: when certain statistics can be computed without labels, the overall label budget shrinks.

Lower bounds and minimax theory. On the information-theoretic side, Castro and Nowak [18] applied mutual information arguments to establish fundamental limits on active learning. More broadly, Assouad’s method, Fano’s inequality, and Le Cam’s two-point method [26, 66, 73] provide the core toolkit for minimax lower bounds in statistics and learning theory. Yang [71] applied these techniques to nonparametric classification; Duchi [35] gave modern treatments connecting information theory to statistical minimax theory; and Agarwal et al. [2] established information-theoretic limits for ranking and selection problems. Our lower-bound constructions adapt the multi-hypothesis testing framework from this line of work: we construct k alternative distributions that an algorithm must distinguish, and bound the total variation between their induced sample distributions using per-query KL divergence.

Practical active learning and structured queries. Adaptive query strategies with provable guarantees have been extended to structured settings, including submodular function optimization [43] and interactive hypothesis testing [21]. In practice, Query-by-Committee [42] and uncertainty sampling [24, 60] concentrate label queries near the current decision boundary. While efficient for accuracy, these heuristics can be blind to minority groups whose probability mass lies far from the boundary—precisely the scenario where our stratified sampling approach provides the strongest advantage.

A.2 Group Fairness: Definitions, Impossibilities, and Algorithms

Fairness definitions. Demographic Parity (DP) requires that positive prediction rates be equal across protected groups [16, 41, 52]. Hardt et al. [48] introduced Equal Opportunity (EO), which equalizes true positive rates $\Pr[h(X)=1 \mid Y=1, A=a]$ across groups, and Equalized Odds (EqOdds), which additionally constrains false positive rates $\Pr[h(X)=1 \mid Y=0, A=a]$. This DP–EO–EqOdds progression—from unconditional to $Y=1$ -conditional to fully Y -conditional constraints—directly shapes our complexity hierarchy: each additional conditioning event introduces an irreducible label cost because the relevant conditional distribution can only be estimated from labeled examples.

Dwork et al. [37] formalized individual fairness, requiring that similar individuals receive similar outcomes under a task-specific similarity metric. Yona and Rothblum [72] gave PAC-style learnability guarantees for metric fairness with finite-sample bounds. Verma and Rubin [68] catalogued over twenty fairness definitions, highlighting the diversity of criteria and the challenge of simultaneously satisfying multiple notions.

Impossibility results and accuracy–fairness tradeoffs. A recurring theme is that different fairness notions conflict with each other and with accuracy. Kleinberg et al. [55] and Chouldechova [22] proved that calibration and balance (a notion closely related to EqOdds) are

jointly satisfiable only when group base rates coincide—an impossibility that motivates the approximate, γ -tolerance formulation we adopt throughout.

The accuracy cost of fairness constraints has been characterized from multiple angles. Corbett-Davies et al. [25] analyzed the accuracy cost of DP constraints in criminal justice, showing that threshold-based fairness can be suboptimal. Menon and Williamson [59] provided a precise characterization of the cost of fairness in binary classification, deriving Bayes-optimal classifiers under DP and EO constraints and quantifying the excess risk incurred by each. Their results are complementary to ours: while they characterize the *statistical* cost of fairness (how much accuracy decreases for a fixed dataset), we characterize the *label* cost (how many additional labels are needed to verify and enforce fairness). Zhao and Gordon [77] proved inherent tradeoffs in fair representations, showing that no representation can simultaneously preserve accuracy and guarantee DP. Woodworth et al. [69] characterized the statistical cost of non-discriminatory prediction in terms of the divergence between group-conditional distributions.

Liu et al. [57] showed that unconstrained learning can itself satisfy implicit fairness criteria under certain distributional conditions, suggesting that fairness constraints are not always necessary. In contrast, Liu et al. [56] demonstrated that static fairness constraints can cause unintended long-term harm through feedback loops, underscoring the need for careful constraint design. Dwork and Ilvento [38] studied how fairness guarantees degrade under composition—a concern that extends to multi-phase algorithms like ours, which use separate phases for fairness evaluation and accuracy training.

Fair classification algorithms. On the algorithmic side, Agarwal et al. [3] reduced fair classification to a sequence of cost-sensitive learning problems, providing a practical framework with convergence guarantees that we use in our constrained ERM step. Zafar et al. [74, 75] formulated convex proxy constraints for DP and EO, enabling efficient gradient-based optimization. Celis et al. [19] developed meta-algorithms with provable guarantees that can enforce arbitrary linear fairness constraints.

Zemel et al. [76] pioneered the representation-learning approach to fairness, learning intermediate representations that are informative for prediction but uninformative about group membership. Dullerud et al. [36] later exposed persistent subgroup gaps in deep metric learning, showing that representation-based approaches may not eliminate unfairness at finer granularities. Kearns et al. [53] developed subgroup fairness auditing algorithms that can detect violations over exponentially many subgroups, and Hébert-Johnson et al. [51] introduced multicalibration, a richer fairness desideratum that requires calibrated predictions for every identifiable subgroup. Comprehensive surveys appear in Mehrabi et al. [58] and Barocas et al. [12]; Suriyakumar et al. [64] discuss emerging challenges at the intersection of fairness and generative AI. A common thread through all of this work: labeled data is assumed to be available in sufficient quantity, and the focus is on how to *use* that data fairly. Our work addresses the complementary question of how to *acquire* data efficiently when fairness is required.

A.3 Fair Data Acquisition

Theoretical foundations. A growing line of work questions the assumption that labeled data is given. Chen et al. [20] showed that selection bias during data collection drives classifier discrimination even when training algorithms explicitly enforce fairness constraints. This result establishes that *how* labels are acquired matters as much as how they are used—the central premise of our work. Blum and Stangl [15] analyzed conditions under which recovery from biased training data is possible, providing sample complexity bounds for debiasing. Benade et al. [13] studied dynamic fair division under partial information, a resource-allocation perspective on fairness under uncertainty. Chouldechova and Roth [23] explicitly identified a rigorous theoretical foundation for fair data collection as an open problem, noting the gap between the rich algorithmic fairness literature and the limited understanding of data acquisition costs. Our work directly addresses this open problem by providing the first information-theoretic lower bounds and matching algorithms for group-fair active learning.

Practical fair active learning. Several recent methods attack the fair data acquisition problem from a practical angle. Anahideh et al. [4] modified uncertainty sampling to account for group balance, demonstrating empirical gains on tabular benchmarks by oversampling underrepresented groups near the decision boundary. Suh et al. [63] studied the label complexity of mixed-initiative classifier training, where human annotators and the learning algorithm collaborate during the labeling process. Their work shows that the interaction paradigm between human and model significantly affects both sample efficiency and the distribution of labels across subpopulations—a finding that reinforces the importance of principled acquisition design over ad hoc query strategies. Abernethy et al. [1] studied active sampling under a min-max fairness objective in a bandit setting, proving sublinear regret bounds that guarantee no group is systematically under-served. Tae et al. [65] (Falcon) formulated fair active selection as a multi-armed bandit problem, balancing exploration across groups with exploitation of informative instances. Fajri et al. [40] (FAL-CUR) combined clustering-based representativeness with uncertainty sampling to balance groups, using a composite acquisition function that targets both accuracy and demographic coverage. Camilleri et al. [17] analyzed fair active learning in low-data regimes, providing finite-sample bounds under distributional assumptions.

Theoretical fair active learning. On the theory side, Shen et al. [62] proved label complexity bounds for *metric*-fair active learning, where similar individuals must receive similar predictions. Their setting is orthogonal to the group-fairness notions (DP, EO, EqOdds) studied here: metric fairness constrains individual-level predictions based on a similarity function, while group fairness constrains aggregate statistics across protected groups. The structural mechanisms driving label complexity are correspondingly different—metric fairness requires dense sampling near similarity boundaries, while group fairness (as we show) requires sufficient positive examples *within each group* to estimate conditional rates.

B Complete Proofs

B.1 Proof of Lemma 1: Accuracy Lower Bound

PROOF. The $\Omega(d/\varepsilon^2)$ lower bound for agnostic learning with VC dimension d is classical [39, 61]. We include a self-contained proof for completeness.

Distribution family. Let $\{x_1, \dots, x_d\}$ be shattered by $\mathcal{H}$ and let $x_0 \notin \{x_1, \dots, x_d\}$ be a *background point*. For each sign vector $s \in \{-1, +1\}^d$, define a distribution P_s as follows.

Marginal of X . $\Pr[X=x_0] = 1 - d\varepsilon^2$ and $\Pr[X=x_j] = \varepsilon^2$ for $j = 1, \dots, d$. (The constraint $d\varepsilon^2 \leq 1$ holds since $\varepsilon < p_+/4 \leq 1/8$ and $d\varepsilon^2 \leq d/(64) \leq 1$ for reasonable d ; if needed, replace ε^2 by $\min(\varepsilon^2, 1/(2d))$.)

Conditional labels.

$$\Pr_s[Y=1 \mid X=x_0] = p_+, \quad (1)$$

$$\Pr_s[Y=1 \mid X=x_j] = \frac{1+s_j}{2}, \quad j = 1, \dots, d. \quad (2)$$

Labels at signal points are deterministic: $Y = 1[s_j = +1]$.

Bayes error. The Bayes-optimal prediction at x_j is $h_s^*(x_j) = 1[s_j = +1]$, incurring zero error at every signal point. At x_0 , the optimal prediction is $1[p_+ \geq 1/2]$ with error $\min(p_+, 1-p_+)$. Thus

$$\text{opt}_s = \min(p_+, 1-p_+) (1 - d\varepsilon^2) \leq \frac{1}{2}. \quad (3)$$

The hypothesis h_s^* that agrees with the Bayes-optimal rule on all signal points exists in $\mathcal{H}$ (by shattering) and achieves $\text{err}(h_s^*) = \text{opt}_s$. In particular, classifiers with error $\leq \text{opt}_s + \varepsilon$ (and hence $\leq \varepsilon$ when opt_s is small) exist, so the learning task is feasible.

Excess-error analysis. Any hypothesis h that disagrees with h_s^* on signal point x_j incurs excess error ε^2 at that point. To achieve $\text{err}(h) \leq \text{opt}_s + \varepsilon$, at most $\varepsilon/\varepsilon^2 = 1/\varepsilon$ signal points may be misclassified. Hence h must correctly identify s_j on at least $d - 1/\varepsilon$ coordinates. For $d \gg 1/\varepsilon$, this is $\Omega(d)$ coordinates.

Info. per sample. Because the marginal of X is identical across all P_s (it does not depend on s), the query strategy of any adaptive learner is independent of s . Each query returns $X=x_j$ with probability ε^2 , so in n total queries, signal point x_j is observed $N_j \sim \text{Bin}(n, \varepsilon^2)$ times.

A single observation of x_j reveals s_j with certainty (since labels are deterministic), so coordinate j is identified iff $N_j \geq 1$. We need $\Pr[N_j \geq 1] \geq 1 - 1/(3d)$ simultaneously for $\geq 2d/3$ coordinates. Since $\Pr[N_j=0] = (1-\varepsilon^2)^n \leq e^{-n\varepsilon^2}$, we need $e^{-n\varepsilon^2} \leq 1/(3d)$, giving $n \geq \varepsilon^{-2} \ln(3d) = \Omega(d/\varepsilon^2)$.

Formal bound via Assouad's lemma. To apply Assouad's lemma in standard form, soften the labels to $\Pr_s[Y=1 \mid X=x_j] = (1 + s_j(1-2v))/2$ for $v = 1/(4d)$. Then $D_{\text{KL}}(P_s \| P_{s'}) = \varepsilon^2 \cdot D_{\text{KL}}(\text{Ber}(1-v) \| \text{Ber}(v)) = O(\varepsilon^2 \ln d)$ for s, s' differing in one coordinate. Assouad's lemma over the hypercube $\{-1, +1\}^d$ with Hamming loss $\rho(s, s') = |\{j : s_j \neq s'_j\}|/d$ and per-coordinate risk ε^2 yields $n = \Omega(d/(\varepsilon^2 \ln d))$. The $\ln d$ factor is removable by Fano's inequality with the full cube, recovering $n = \Omega(d/\varepsilon^2)$. $\square$

B.2 Proof of Lemma 2: DP Verification

PROOF. **Step 1: Reduction to VC uniform convergence.** The rate $r_a(h)$ depends only on h and the feature X , not on the label Y . Given m unlabeled samples $\{x_1, \dots, x_m\}$ drawn i.i.d. from $\mathcal{D}(X \mid A = a)$, the empirical estimate is:

$$\hat{r}_a(h) = \frac{1}{m} \sum_{i=1}^m 1[h(x_i) = 1].$$

The function class $\mathcal{G} = \{x \mapsto 1[h(x) = 1] : h \in \mathcal{H}\}$ maps $\mathcal{X} \rightarrow \{0, 1\}$ and has VC dimension at most d (equal to that of $\mathcal{H}$, since the indicator does not increase shattering capacity).

Step 2: Uniform convergence bound. By the VC inequality (see, e.g., Theorem 6.8 of Shalev-Shwartz and Ben-David 61), for a function class of VC dimension d , m i.i.d. samples, and any $t > 0$:

$$\Pr \left[\sup_{h \in \mathcal{H}} |\hat{r}_a(h) - r_a(h)| > t \right] \leq 8 \cdot \left(\frac{em}{d} \right)^d \exp(-mt^2/8).$$

Set $t = \gamma/8$ and the failure probability to $\delta' = \delta/(2k)$ (to allow a union bound over k groups and additional events). We need:

$$8 \cdot \left(\frac{em}{d} \right)^d \exp(-m\gamma^2/512) \leq \frac{\delta}{2k}.$$

Taking logarithms:

$$m \cdot \frac{\gamma^2}{512} \geq \ln \frac{16k}{\delta} + d \ln \frac{em}{d}.$$

Setting $m = C(d \ln(d/\gamma) + \ln(k/\delta))/\gamma^2$ for a sufficiently large constant C satisfies this inequality: the term $d \ln(em/d)$ evaluates to $O(d \ln(d/\gamma))$ for fixed C , and $\ln(16k/\delta)$ is absorbed by $C \ln(k/\delta)/\gamma^2 \cdot \gamma^2/512$.

Step 3: Union bound over groups. Applying the bound independently to each group $a \in [k]$ with failure probability $\delta/(2k)$ per group:

$$\Pr \left[\exists a \in [k] : \sup_{h \in \mathcal{H}} |\hat{r}_a(h) - r_a(h)| > \frac{\gamma}{8} \right] \leq k \cdot \frac{\delta}{2k} = \frac{\delta}{2}.$$

This establishes the stated sample complexity per group with overall failure probability at most $\delta/2$, leaving $\delta/2$ for other events in the main algorithm.

Implication for DP verification. For any classifier h :

$$\text{DP}(h) = \max_{a, a' \in [k]} |r_a(h) - r_{a'}(h)|.$$

With $|\hat{r}_a(h) - r_a(h)| \leq \gamma/8$ for all a and h :

$$|\widehat{\text{DP}}(h) - \text{DP}(h)| \leq 2 \cdot \frac{\gamma}{8} = \frac{\gamma}{4}.$$

No labels are required at any stage. □

B.3 Proof of Lemma 3: TPR Verification Lower Bound

We first introduce notation and a supporting lemma used throughout this proof and Appendix B.4.

Notation. The *total variation distance* between two probability distributions P and Q on the same measurable space is $\text{TV}(P, Q) = \sup_E |P(E) - Q(E)|$, where the supremum is over all measurable events E . The *Kullback–Leibler (KL) divergence* from P to Q is $D_{\text{KL}}(P \| Q) = \int \ln(dP/dQ) dP$ when $P \ll Q$. *Pinsker's inequality* states $\text{TV}(P, Q) \leq \sqrt{D_{\text{KL}}(P \| Q)}/2$.

LEMMA 4 (CHAIN RULE FOR ADAPTIVE QUERIES). *Let P and Q be two joint distributions over a sequence $(Z_1, \dots, Z_n)$ generated by an adaptive process, where each Z_i may depend on $Z_{<i} = (Z_1, \dots, Z_{i-1})$. Then:*

$$D_{\text{KL}}(P^n \| Q^n) = \sum_{i=1}^n \mathbb{E}_P [D_{\text{KL}}(P(Z_i | Z_{<i}) \| Q(Z_i | Z_{<i}))].$$

PROOF. By the chain rule of KL divergence applied to the factorization $P(Z_1, \dots, Z_n) = \prod_{i=1}^n P(Z_i | Z_{<i})$, which holds for any joint distribution including adaptive processes. □

We also record the following standard bound on the KL divergence between Bernoulli distributions:

$$D_{\text{KL}}(\text{Ber}(p) \| \text{Ber}(p+\delta)) \leq \frac{\delta^2}{p(1-p) - \delta^2}, \quad \text{for } |\delta| < \min(p, 1-p). \quad (4)$$

PROOF OF LEMMA 3. We reduce to a hypothesis testing problem. The key insight is that *even when accuracy is trivially achievable*, certifying $\text{EO} \leq \gamma$ requires estimating each group's TPR to precision γ , which demands many labeled positives.

Step 1: Construction. Let $\mathcal{H} = \{h_t(x) = \mathbf{1}[x \geq t] : t \in [0, 1]\}$, $X \sim \text{Unif}[0, 1]$, and $\pi_a = 1/k$ for all $a \in [k]$. Fix the optimal threshold $t^* = 1 - p_+$ (so $\Pr[X \geq t^*] = p_+$) and a noise parameter $\eta > 0$ to be chosen. For a group-specific perturbation vector $\beta = (\beta_1, \dots, \beta_k)$, define:

$$\Pr[Y=1 | X, A=a] = \begin{cases} 1 - \eta + \beta_a & \text{if } X \geq t^*, \\ \eta - \frac{p_+}{1-p_+} \beta_a & \text{if } X < t^*. \end{cases} \quad (5)$$

Step 2: Marginal preservation. For every group a :

$$\begin{aligned} \Pr[Y=1 | A=a] &= p_+ (1 - \eta + \beta_a) + (1-p_+) \left(\eta - \frac{p_+ \beta_a}{1-p_+} \right) \\ &= p_+ (1-\eta) + p_+ \beta_a + (1-p_+) \eta - p_+ \beta_a = p_+. \end{aligned}$$

The β_a terms cancel, so $\Pr[Y=1 | A=a] = p_+$ regardless of β_a .

Step 3: Hypotheses.

- H_0 : $\beta_a = 0$ for all $a \in [k]$.
- $H_1(a^*)$: $\beta_{a^*} = 2\gamma$, $\beta_a = 0$ for $a \neq a^*$, with $a^* \sim \text{Unif}([k])$.

Step 4: Validity conditions. We need all probabilities in (5) to lie in $[0, 1]$. The binding constraint is the lower region for group a^* :

$$\eta - \frac{p_+ \cdot 2\gamma}{1 - p_+} \geq 0 \iff \eta \geq \frac{2\gamma p_+}{1 - p_+}.$$

For the upper region: $1 - \eta + 2\gamma \leq 1$ iff $\eta \geq 2\gamma$. Both are satisfied by choosing η appropriately:

$$\eta = \max\left(\frac{\varepsilon}{4}, \frac{2\gamma p_+}{1 - p_+}, 2\gamma\right). \quad (6)$$

Since $\gamma \leq 1/8$ and $p_+ \leq 1/2$: $\frac{2\gamma p_+}{1 - p_+} \leq \frac{2 \cdot (1/8) \cdot (1/2)}{1/2} = 1/4$, and $2\gamma \leq 1/4$. Also $\varepsilon/4 \leq p_+/16 \leq 1/32$. So $\eta \leq 1/4 < 1$, and all probabilities are valid.

Step 5: Feasibility. The classifier h_{t^*} achieves:

$$\begin{aligned} \text{err}_a(h_{t^*}) &= p_+ \Pr[Y=0 \mid X \geq t^*, A=a] + (1-p_+) \Pr[Y=1 \mid X < t^*, A=a] \\ &= p_+(\eta - \beta_a) + (1-p_+)\left(\eta - \frac{p_+\beta_a}{1-p_+}\right) = \eta - 2p_+\beta_a. \end{aligned} \quad (7)$$

Under H_0 : $\text{err}(h_{t^*}) = \eta$. Under $H_1(a^*)$: $\text{err}(h_{t^*}) = \eta - 4p_+\gamma/k < \eta$.

When $\eta = \varepsilon/4$: $\text{err}(h_{t^*}) \leq \varepsilon/4 < \varepsilon$. When $\eta > \varepsilon/4$ (i.e., the other terms in (6) dominate): $\text{err}(h_{t^*}) = \eta \leq \max(2\gamma, 2\gamma p_+/(1-p_+))$. We require $\eta \leq \varepsilon$ for feasibility; see Step 9 for the regime analysis.

In all valid regimes, h_{t^*} achieves $\text{err}(h_{t^*}) \leq \varepsilon$, so a PAC-fair solution exists.

Step 6: True positive rate.

$$\text{TPR}_a(h_{t^*}) = \Pr[X \geq t^* \mid Y=1, A=a] = \frac{(1-\eta+\beta_a)p_+}{p_+} = 1 - \eta + \beta_a. \quad (8)$$

Under H_0 : $\text{TPR}_a = 1 - \eta$ for all a , so $\text{EO}(h_{t^*}) = 0$.

Under $H_1(a^*)$: $\text{TPR}_{a^*} = 1 - \eta + 2\gamma$, others $= 1 - \eta$, so $\text{EO}(h_{t^*}) = 2\gamma > \gamma$. Thus h_{t^*} violates the fairness constraint under H_1 , and any PAC-fair learner must distinguish H_0 from H_1 (or find a different classifier).

Step 7: Testing reduction. Let $\bar{P}_1 = k^{-1} \sum_{a=1}^k P_{1,a}$ be the mixture under H_1 . Any algorithm that produces a classifier with $\text{EO}(h) \leq \gamma$ on both H_0 and H_1 with probability $\geq 2/3$ must satisfy $\text{TV}(P_0^n, \bar{P}_1^n) \geq 1/3$.

By the convexity of KL divergence: $D_{\text{KL}}(P_0^n \parallel \bar{P}_1^n) \leq k^{-1} \sum_{a=1}^k D_{\text{KL}}(P_0^n \parallel P_{1,a}^n)$. For fixed anomalous group a , only queries with $A_i = a$ contribute. By the chain rule for adaptive queries: $D_{\text{KL}}(P_0^n \parallel P_{1,a}^n) \leq n_a \cdot D_a$, where $n_a = \mathbb{E}_{P_0}[\# \text{ queries with } A_i = a] = n/k$ (by symmetry) and D_a is the per-query divergence.

Step 8: KL divergence per query. For a query to group a^* , conditioning on X :

Case $X \geq t^$ (prob. p_+):* $H_0: Y \sim \text{Ber}(1-\eta)$. $H_1: Y \sim \text{Ber}(1-\eta+2\gamma)$. Using $D_{\text{KL}}(\text{Ber}(p) \parallel \text{Ber}(p+\delta)) \leq \delta^2/(p(1-p) - \delta^2)$ with $p = 1-\eta$, $\delta = 2\gamma$:

$$D_{\text{upper}} \leq \frac{4\gamma^2}{(1-\eta)\eta - 4\gamma^2}. \quad (9)$$

Since $\eta \geq 2\gamma$: $(1-\eta)\eta \geq \eta/2 \geq \gamma$ and $4\gamma^2 \leq \eta\gamma/2 \leq (1-\eta)\eta/4$. Thus the denominator $\geq 3(1-\eta)\eta/4 \geq 3\eta/8$, giving

$$D_{\text{upper}} \leq \frac{32\gamma^2}{3\eta}. \quad (10)$$

Case $X < t^$ (prob. $1-p_+$):* $H_0: Y \sim \text{Ber}(\eta)$. $H_1: Y \sim \text{Ber}(\eta - 2\gamma p_+/(1-p_+))$. With $\delta = 2\gamma p_+/(1-p_+)$:

$$D_{\text{lower}} \leq \frac{4\gamma^2 p_+^2 / (1-p_+)^2}{\eta(1-\eta) - 4\gamma^2 p_+^2 / (1-p_+)^2}. \quad (11)$$

By the same argument (the perturbation is smaller than in the upper regime since $p_+/(1-p_+) \leq 1$),

$$D_{\text{lower}} \leq \frac{32\gamma^2 p_+^2}{3\eta(1-p_+)^2}. \quad (12)$$

Expected KL per query to group a^ :*

$$\begin{aligned} D_{a^*} &= p_+ \cdot D_{\text{upper}} + (1-p_+) \cdot D_{\text{lower}} \\ &\leq \frac{32p_+\gamma^2}{3\eta} + \frac{32p_+^2\gamma^2}{3\eta(1-p_+)} = \frac{32p_+\gamma^2}{3\eta(1-p_+)}. \end{aligned} \quad (13)$$

Step 9: Regime analysis and final bound. Substituting η from (6):

Regime I: $\eta = \varepsilon/4$ (i.e., $\varepsilon/4 \geq 2\gamma$ and $\varepsilon/4 \geq 2\gamma p_+/(1-p_+)$).

$$D_{a^*} \leq \frac{128 p_+ \gamma^2}{3 \varepsilon (1-p_+)}.$$

Total KL: $D_{\text{KL}}(P_0^n \| \bar{P}_1^n) \leq \frac{n}{k} \cdot \frac{128 p_+ \gamma^2}{3 \varepsilon (1-p_+)}$. Pinsker:

$$n \geq \frac{3 k \varepsilon (1-p_+)}{1152 p_+ \gamma^2} = \Omega\left(\frac{k \varepsilon}{\gamma^2 p_+}\right). \quad (14)$$

Regime II: $\eta = 2\gamma$ (i.e., $2\gamma > \varepsilon/4$, which means $\gamma > \varepsilon/8$). Feasibility requires $\eta = 2\gamma \leq \varepsilon$, hence $\gamma \leq \varepsilon/2$.

$$D_{a^*} \leq \frac{32 p_+ \gamma^2}{3 \cdot 2\gamma (1-p_+)} = \frac{16 p_+ \gamma}{3(1-p_+)}.$$

Total KL: $\leq \frac{16 p_+ \gamma n}{3 k (1-p_+)}$. Pinsker:

$$n \geq \frac{3 k (1-p_+)}{144 p_+ \gamma} = \Omega\left(\frac{k}{\gamma p_+}\right). \quad (15)$$

Combined bound. Taking the maximum of the two regimes:

$$n \geq \Omega\left(\frac{k}{\gamma^2 p_+} \cdot \min(\varepsilon, \gamma)\right). \quad (16)$$

When $\varepsilon \geq \gamma$ (which holds whenever $\gamma \leq \varepsilon$, i.e., γ is the binding constraint): the bound is $\Omega(k/(\gamma p_+))$. When $\varepsilon < \gamma$: the bound is $\Omega(k\varepsilon/(\gamma^2 p_+))$.

REMARK 2 (RECOVERING $\Omega(k/(\gamma^2 p_+))$ UNDER EXCESS RISK). *The factor of $\min(\varepsilon, \gamma)$ arises because the noise parameter η must satisfy $\eta \leq \varepsilon$ for the absolute-error requirement $\text{err}(h) \leq \varepsilon$. Under the excess-risk formulation $\text{err}(h) - \text{opt}_{\mathcal{H}} \leq \varepsilon$, the noise parameter is unconstrained. Setting $\eta = 1/4$ gives $(1-\eta)\eta = 3/16 = \Theta(1)$, hence $D_{a^*} = O(\gamma^2 p_+)$ and the bound becomes $n \geq \Omega(k/(\gamma^2 p_+))$, matching the upper bound of Theorem 3 up to logarithmic factors.*

□

B.4 Proof of Lemma 5: FPR Lower Bound

LEMMA 5 (FPR VERIFICATION LOWER BOUND). *Let $0 < p_+ \leq 1/2$ and $\gamma \leq 1/8$. Any algorithm that achieves $\text{err}(h) \leq \varepsilon$ and certifies $\max_{a,a'} |\text{FPR}_a(h) - \text{FPR}_{a'}(h)| \leq \gamma$ with probability $\geq 2/3$ requires $\Omega(k/(\gamma^2 p_-))$ labeled examples, where $p_- = 1 - p_+$.*

PROOF. The proof mirrors Lemma 3 with the roles of $Y=1$ and $Y=0$ exchanged.

Construction. Use the same setup as Appendix B.3 with threshold $t^* = p_-$ (so $\Pr[X < t^*] = p_- = 1 - p_+$), noise parameter

$$\eta = \max\left(\frac{\varepsilon}{4}, \frac{2\gamma p_-}{1-p_-}, 2\gamma\right) = \max\left(\frac{\varepsilon}{4}, \frac{2\gamma p_-}{p_+}, 2\gamma\right), \quad (17)$$

and perturbations affecting the *negative*-class conditional:

$$\Pr[Y=0 \mid X, A=a] = \begin{cases} 1 - \eta + \beta_a & \text{if } X < t^*, \\ \eta - \frac{p_-}{1-p_-} \beta_a & \text{if } X \geq t^*. \end{cases}$$

Under H_0 : $\beta_a = 0$ for all a . Under $H_1(a^*)$: $\beta_{a^*} = 2\gamma$, others 0.

Validity conditions. The binding constraint for the upper region ($X \geq t^*$, group a^*) is:

$$\eta - \frac{p_- \cdot 2\gamma}{1-p_-} = \eta - \frac{2\gamma p_-}{p_+} \geq 0,$$

which holds by the choice of η in (17). Note that since $p_- \geq 1/2 \geq p_+$, the ratio p_-/p_+ can exceed 1, making this term potentially larger than 2γ ; this is why the middle term in (17) is essential and cannot be omitted (unlike the TPR case where $p_+/(1-p_+) \leq 1$).

Since $\gamma \leq 1/8$ and $p_- \leq 1$: $2\gamma p_-/p_+ \leq 1/(4p_+)$. For $p_+ \geq 1/4$ (the typical regime) this is at most 1, well within $[0, 1]$. For very small p_+ , this term dominates η , but the resulting bound remains valid.

Key properties. $\Pr[Y=0 \mid A=a] = p_-$ is preserved (by an identical cancellation argument). $\text{err}(h_{t^*}) = \eta \leq \varepsilon$ (in all valid regimes, as in the TPR proof). $\text{FPR}_a(h_{t^*}) = 1 - \eta + \beta_a$. Under H_1 : FPR gap $= 2\gamma > \gamma$.

Lower bound. By the same KL analysis as Lemma 3 with p_+ replaced by p_- throughout:

$$n \geq \Omega\left(\frac{k}{\gamma^2 p_-} \cdot \min(\epsilon, \gamma)\right). \quad (18)$$

Under the excess-risk formulation, this becomes $\Omega(k/(\gamma^2 p_-))$. $\square$

B.5 Proof of Theorem 2: Separation

We recall the *disagreement coefficient* [47]. For a hypothesis class $\mathcal{H}$, target $h^* \in \mathcal{H}$, and radius $r > 0$, let $B(h^*, r) = \{h \in \mathcal{H} : \Pr[h(X) \neq h^*(X)] \leq r\}$ denote the ball of hypotheses within error r of h^* , and $\text{DIS}(V) = \{x \in \mathcal{X} : \exists h, h' \in V, h(x) \neq h'(x)\}$ the disagreement region of a set $V \subseteq \mathcal{H}$. The disagreement coefficient is $\theta = \sup_{r>0} \Pr[X \in \text{DIS}(B(h^*, r))] / r$.

PROOF. The proof combines two ingredients: the well-known logarithmic label complexity of unconstrained active learning for certain hypothesis classes, and the lower bound from Lemma 3.

Part 1: Unconstrained active learning is fast. Consider the class of threshold classifiers $\mathcal{H} = \{h_t : t \in [0, 1]\}$ on $\mathcal{X} = [0, 1]$ with $h_t(x) = \mathbf{1}[x \geq t]$. In the realizable setting, the disagreement coefficient θ for threshold classifiers under any continuous marginal distribution on $[0, 1]$ is $\theta = O(1)$ [47]. Specifically, for the uniform distribution, the disagreement region around the true threshold t^* at radius r has measure $\Pr[X \in \text{DIS}(B(h^*, r))] \leq 2r$, giving $\theta \leq 2$.

By the disagreement-based active learning framework of Dasgupta et al. [29], Hanneke [47], the label complexity is:

$$n_{\text{unconstrained}} = O(\theta d \log(1/\epsilon)) = O(\log(1/\epsilon)),$$

since $d = 1$ for threshold classifiers and $\theta = O(1)$.

Part 2: EO-constrained learning is slow. The lower bound from Lemma 3 applies to *any* algorithm that simultaneously achieves accuracy ϵ and fairness γ , regardless of whether a perfectly accurate classifier exists. The construction in Appendix B.3 produces distributions where the optimal threshold h_{t^*} exists in $\mathcal{H}$ and achieves error $\eta \leq \epsilon$, while certifying that $\text{EO}(h_{t^*}) \leq \gamma$ requires $\Omega(k/(\gamma^2 p_+) \cdot \min(\epsilon, \gamma))$ labels.

Crucially, this lower bound holds for the *same* hypothesis class $\mathcal{H}$ (thresholds) used in Part 1. The lower bound is a *minimax* statement: there *exist* distributions over (X, Y, A) —namely those in the construction of Lemma 3—for which any EO-constrained algorithm requires $\Omega(k/(\gamma^2 p_+) \cdot \min(\epsilon, \gamma))$ labels.

Part 3: Separation. The separation follows from comparing two regimes: on favorable (realizable) distributions, unconstrained active learning uses $O(\theta d \log(1/\epsilon))$ labels; on adversarial distributions from Lemma 3, EO-constrained learning requires $\Omega(k/(\gamma^2 p_+) \cdot \min(\epsilon, \gamma))$ labels, and since a PAC-fair learner must succeed on *all* distributions, its worst-case label complexity is at least this quantity.

When $k \min(\epsilon, \gamma)/(\gamma^2 p_+) \gg \theta d \log(1/\epsilon)$, the fairness constraint imposes a cost that exceeds the entire unconstrained budget, even in the most favorable (realizable) case for the unconstrained learner. This collapses the exponential $(\log(1/\epsilon))$ speedup of active learning to polynomial rates.

Near-realizable lower bound. To complement the minimax argument, we provide a lower bound for *nearly* realizable distributions (Bayes error γ/k , vanishing as $k \rightarrow \infty$).

Consider two hypotheses:

- H_0 : $\Pr[Y=1 | X \geq t^*] = 1$ and $\Pr[Y=1 | X < t^*] = 0$ for all groups, so h_{t^*} achieves $\text{err} = 0$ and $\text{EO} = 0$.
- $H_1(a^*)$: same as H_0 except $\Pr[Y=1 | X < t^*, A=a^*] = 2\gamma$ for a uniformly random group a^* .

Under $H_1(a^*)$: $\text{err}(h_{t^*}) = \gamma/k$ (small), and $\text{TPR}_{a^*} = 1/(1+2\gamma) < 1$ while $\text{TPR}_a = 1$ for $a \neq a^*$, giving $\text{EO}(h_{t^*}) = 2\gamma/(1+2\gamma) > \gamma/2$.

The per-query KL divergence for group a^* in the region $X < t^*$ is:

$$D_{\text{KL}}(\text{Ber}(0) \parallel \text{Ber}(2\gamma)) = -\ln(1-2\gamma) \leq 4\gamma \quad \text{for } \gamma \leq 1/8.$$

By the same randomization argument as Lemma 3: $D_{\text{KL}}(P_0^n \parallel \bar{P}_1^n) \leq n \cdot 4\gamma/(2k) = 2n\gamma/k$. Setting this to $\Omega(1)$:

$$n \geq \Omega\left(\frac{k}{\gamma}\right).$$

This is a weaker bound than $\Omega(k/(\gamma^2 p_+))$ but holds strictly within the nearly realizable setting. Since $k/\gamma \gg \log(1/\epsilon)$ in the regime of interest, the qualitative separation—conditional fairness eliminating exponential speedups—holds in either case.

REMARK 3. Under $H_1(a^*)$, $\text{opt}_{\mathcal{H}} = \gamma/k > 0$, so the instance is not strictly realizable. However, for $k \geq \gamma/\epsilon$ we have $\gamma/k \leq \epsilon$, and the classifier h_{t^*} satisfies $\text{err}(h_{t^*}) \leq \epsilon$. The separation $\Omega(k/\gamma) \gg O(\log(1/\epsilon))$ holds in either case. $\square$

DP-fair learning retains exponential speedups because DP verification is label-free, reinforcing the structural dichotomy at the heart of our theory.

B.6 Proof of Theorem 3: Upper Bounds

Throughout this section, we use Assumption 1: there exists $h^* \in \mathcal{H}$ with $\text{err}(h^*) \leq \varepsilon/4$ and $\mathcal{F}(h^*) \leq \gamma/8$. We also recall that $p_a = \Pr[Y=1 \mid A=a]$ denotes the group-specific base rate (Section 3). The constants c_1, c_2 below are absolute constants arising from the VC uniform convergence theorem [61, Theorem 6.8] and Lemma 2, respectively.

B.6.1 DP Case: Proof for ConstrainedERM-DP.

PROOF. The algorithm has two phases plus a solve step. *Phase 1 (Labels for accuracy):* Draw $n_L = c_1 d \varepsilon^{-2} \ln(4/\delta)$ i.i.d. labeled samples S_L for a constant c_1 from the VC uniform convergence bound. *Phase 2 (Unlabeled for fairness):* For each group a , draw $m_a = c_2 (d \ln(d/\gamma) + \ln(8k/\delta))/\gamma^2$ unlabeled samples and compute $\hat{r}_a(h)$ for all $h \in \mathcal{H}$. *Solve:* $\hat{h} \leftarrow \arg \min_{h: \widehat{\text{DP}}(h) \leq \gamma/2} \widehat{\text{err}}(h; S_L)$.

Event 1: Accuracy estimation. By the VC uniform convergence theorem, with $n_L = c_1 d \varepsilon^{-2} \ln(4/\delta)$:

$$\Pr \left[\sup_{h \in \mathcal{H}} |\widehat{\text{err}}(h; S_L) - \text{err}(h)| > \varepsilon/4 \right] \leq \frac{\delta}{4}. \quad (19)$$

Event 2: Fairness estimation. By Lemma 2 with precision $\gamma/8$ per group:

$$\Pr \left[\exists a : \sup_{h \in \mathcal{H}} |\hat{r}_a(h) - r_a(h)| > \gamma/8 \right] \leq \frac{\delta}{4}. \quad (20)$$

Feasibility of h^* . Conditioned on both events, h^* (from Assumption 1) has:

$$\widehat{\text{DP}}(h^*) \leq \text{DP}(h^*) + 2 \cdot \frac{\gamma}{8} = \frac{\gamma}{8} + \frac{\gamma}{4} = \frac{3\gamma}{8} < \frac{\gamma}{2}.$$

Therefore h^* is feasible for the constrained ERM.

Accuracy guarantee. Since h^* is feasible and $\hat{h}$ minimizes empirical error over the feasible set:

$$\widehat{\text{err}}(\hat{h}; S_L) \leq \widehat{\text{err}}(h^*; S_L) \leq \text{err}(h^*) + \frac{\varepsilon}{4} \leq \frac{\varepsilon}{4} + \frac{\varepsilon}{4} = \frac{\varepsilon}{2}.$$

Therefore: $\text{err}(\hat{h}) \leq \widehat{\text{err}}(\hat{h}) + \varepsilon/4 \leq \varepsilon/2 + \varepsilon/4 = 3\varepsilon/4 < \varepsilon$.

Fairness guarantee. Since $\widehat{\text{DP}}(\hat{h}) \leq \gamma/2$:

$$\text{DP}(\hat{h}) \leq \widehat{\text{DP}}(\hat{h}) + 2 \cdot \frac{\gamma}{8} \leq \frac{\gamma}{2} + \frac{\gamma}{4} = \frac{3\gamma}{4} < \gamma.$$

Union bound. The overall failure probability is at most $\delta/4 + \delta/4 = \delta/2 < \delta$, confirming the $(\varepsilon, \gamma, \delta)$ -PAC guarantee.

Complexity. The label complexity is $n_L = O(d \varepsilon^{-2} \ln(1/\delta))$. The unlabeled cost per group is $O((d \ln(d/\gamma) + \ln(k/\delta))/\gamma^2)$, totaling $O(k(d \ln(d/\gamma) + \ln(k/\delta))/\gamma^2)$. $\square$

B.6.2 EO Case: Proof for FairStratified. We recall two standard distributions from probability theory. The *geometric distribution* $\text{Geom}(p)$ models the number of independent $\text{Ber}(p)$ trials needed to obtain the first success, with $\mathbb{E}[G] = 1/p$ and $\text{Var}(G) = (1-p)/p^2$. The *negative binomial distribution* $\text{NegBin}(r, p)$ models the number of independent $\text{Ber}(p)$ trials needed to accumulate r successes. Equivalently, $T \sim \text{NegBin}(r, p)$ can be written as $T = \sum_{i=1}^r G_i$ where $G_i \stackrel{\text{i.i.d.}}{\sim} \text{Geom}(p)$, giving $\mathbb{E}[T] = r/p$ and $\text{Var}(T) = r(1-p)/p^2$.

PROOF. The algorithm proceeds in two independent phases.

Phase 1: Stratified fairness evaluation. For each group $a \in [k]$, sample instances from group a uniformly at random (from the unlabeled pool), query their labels, and continue until $2n_a^+$ positive instances are collected, where:

$$n_a^+ = \left\lceil \frac{32}{\gamma^2} \ln \frac{8k}{\delta} \right\rceil.$$

Split the $2n_a^+$ positives into two equal halves:

- $S_{\text{ctr}}: n_a^+$ positives per group (constraint set),
- $S_{\text{val}}: n_a^+$ positives per group (validation set).

Let T_a denote the total number of queries to group a (including negatives seen along the way). Then $T_a \sim \text{NegBin}(2n_a^+, p_a)$, where $p_a = \Pr[Y=1 \mid A=a]$.

Phase 2: i.i.d. accuracy training. Independently of Phase 1, draw $n_{\text{train}} = c_1 d \varepsilon^{-2} \ln(4/\delta)$ i.i.d. labeled samples S_{train} .

Constrained ERM. Solve: $\hat{h} \leftarrow \arg \min_{h: \widehat{\text{EO}}(h; S_{\text{ctr}}) \leq \gamma/2} \widehat{\text{err}}(h; S_{\text{train}})$.

We now verify the four required properties.

(A) Fairness guarantee.

Recall from Algorithm 2 that Phase 1 collects $2n_a^+$ positives per group and splits them into two independent halves: S_{ctr} (n_a^+ positives per group, constraint set) and S_{val} (n_a^+ positives per group, validation set). The constrained ERM is $\hat{h} \leftarrow \arg \min_{h: \widehat{\text{EO}}(h; S_{\text{ctr}}) \leq \gamma/2} \widehat{\text{err}}(h; S_{\text{train}})$.

Key observation. $\hat{h}$ depends on S_{cstr} (through the feasible set) and S_{train} (through the objective), but is *independent* of S_{val} .

Step A.1: Constraint-side uniform convergence. Because $\hat{h}$ depends on S_{cstr} , we cannot apply Hoeffding to bound $|\widehat{\text{TPR}}_a(\hat{h}; S_{\text{cstr}}) - \text{TPR}_a(\hat{h})|$ directly. Instead, we use VC uniform convergence.

The function class $\{(x, y) \mapsto \mathbf{1}[h(x)=1, y=1] : h \in \mathcal{H}\}$ has VC dimension at most d . With n_a^+ i.i.d. positives from group a in S_{cstr} , the VC uniform convergence theorem gives: with probability $\geq 1 - \delta/4$,

$$\sup_{h \in \mathcal{H}, a \in [k]} |\widehat{\text{TPR}}_a(h; S_{\text{cstr}}) - \text{TPR}_a(h)| \leq \gamma/4, \quad (21)$$

provided $n_a^+ \geq c_0(d \log(d/\gamma) + \log(k/\delta))/\gamma^2$ for a universal constant c_0 . (This is satisfied by our choice $n_a^+ = (32/\gamma^2) \ln(8k/\delta)$ when d is absorbed into the logarithmic factor in Theorem 3's statement.)

On this event, the constraint $\widehat{\text{EO}}(\hat{h}; S_{\text{cstr}}) \leq \gamma/2$ implies:

$$\text{EO}(\hat{h}) \leq \gamma/2 + 2 \cdot \gamma/4 = \gamma.$$

This is the main fairness guarantee: $\text{EO}(\hat{h}) \leq \gamma$.

Step A.2: Validation-side concentration (independent certificate). Since $\hat{h} \perp S_{\text{val}}$, we may treat $\hat{h}$ as a *fixed* classifier. The n_a^+ positives in $S_{\text{val},a}$ are i.i.d. from $\mathcal{D}(X | Y=1, A=a)$. By Hoeffding's inequality with $t = \gamma/8$ and $n_a^+ = (32/\gamma^2) \ln(8k/\delta)$:

$$\Pr[|\widehat{\text{TPR}}_a(\hat{h}; S_{\text{val}}) - \text{TPR}_a(\hat{h})| > \gamma/8] \leq 2 \exp(-2n_a^+ \gamma^2/64) = \frac{\delta}{4k}.$$

A union bound over k groups:

$$\Pr[\exists a : |\widehat{\text{TPR}}_a(\hat{h}; S_{\text{val}}) - \text{TPR}_a(\hat{h})| > \gamma/8] \leq \frac{\delta}{4}. \quad (22)$$

The validation set provides an independent data-dependent certificate of the fairness level, but the theoretical guarantee $\text{EO}(\hat{h}) \leq \gamma$ already follows from Step A.1 alone.

Step A.3: Combining. On the event (21) (probability $\geq 1 - \delta/4$), Step A.1 guarantees $\text{EO}(\hat{h}) \leq \gamma$.

(B) Feasibility of h^* .

By Assumption 1, $\text{EO}(h^*) \leq \gamma/8$. Because h^* is a fixed hypothesis independent of S_{cstr} , Hoeffding's inequality on the n_a^+ i.i.d. positives in each $S_{\text{cstr},a}$ gives:

$$\widehat{\text{EO}}(h^*; S_{\text{cstr}}) \leq \text{EO}(h^*) + 2 \cdot \frac{\gamma}{8} = \frac{\gamma}{8} + \frac{\gamma}{4} = \frac{3\gamma}{8} < \frac{\gamma}{2},$$

with probability $\geq 1 - \delta/4$. Therefore h^* is feasible for the constrained ERM.

(C) Accuracy guarantee.

Since S_{train} is i.i.d., the VC uniform convergence theorem gives:

$$\Pr\left[\sup_{h \in \mathcal{H}} |\widehat{\text{err}}(h; S_{\text{train}}) - \text{err}(h)| > \varepsilon/4\right] \leq \frac{\delta}{4}. \quad (23)$$

With h^* feasible and $\text{err}(h^*) \leq \varepsilon/4$:

$$\text{err}(\hat{h}) \leq \widehat{\text{err}}(\hat{h}) + \frac{\varepsilon}{4} \leq \widehat{\text{err}}(h^*) + \frac{\varepsilon}{4} \leq \text{err}(h^*) + \frac{\varepsilon}{2} \leq \frac{\varepsilon}{4} + \frac{\varepsilon}{2} = \frac{3\varepsilon}{4} < \varepsilon.$$

(D) Label complexity and termination.

Phase 1 collects $2n_a^+$ positives per group (split into S_{cstr} and S_{val} , each with n_a^+ positives). The number of labels queried for group a is $T_a \sim \text{NegBin}(2n_a^+, p_a)$ with $\mathbb{E}[T_a] = 2n_a^+/p_a$. By the multiplicative Chernoff bound for the negative binomial distribution:

$$\Pr\left[T_a > \frac{4n_a^+}{p_a}\right] \leq (e/4)^{2n_a^+} \leq \exp(-n_a^+/4) \leq \frac{\delta}{4k}. \quad (24)$$

Justification of (24). Write $T_a = \sum_{i=1}^{2n_a^+} G_i$ where $G_i \sim \text{Geom}(p_a)$ are independent geometric random variables (number of trials until the i -th success). Each G_i has mean $1/p_a$ and moment generating function $\mathbb{E}[e^{tG_i}] = p_a e^t / (1 - (1 - p_a)e^t)$ for $t < -\ln(1 - p_a)$.

By the standard multiplicative Chernoff bound, for $\delta_{\text{mult}} > 0$:

$$\Pr[T_a > (1 + \delta_{\text{mult}})\mathbb{E}[T_a]] \leq \left(\frac{e^{\delta_{\text{mult}}}}{(1 + \delta_{\text{mult}})^{1+\delta_{\text{mult}}}} \right)^{2n_a^+}.$$

Setting $\delta_{\text{mult}} = 1$ (i.e., $T_a > 2\mathbb{E}[T_a] = 4n_a^+/p_a$):

$$\Pr\left[T_a > \frac{4n_a^+}{p_a}\right] \leq \left(\frac{e}{4}\right)^{2n_a^+} \leq \exp\left(-\frac{n_a^+}{4}\right),$$

since $\ln(e/4) = 1 - 2 \ln 2 \approx -0.386 < -1/8$ and $2 \cdot 0.386 > 1/4$. ◇

With $n_a^+ = (32/\gamma^2) \ln(8k/\delta) \geq 32 \ln(8k/\delta)$ (since $\gamma \leq 1$):

$$\exp\left(-\frac{n_a^+}{4}\right) \leq \exp(-8 \ln(8k/\delta)) = \left(\frac{\delta}{8k}\right)^8 \leq \frac{\delta}{4k}.$$

Union over k groups: $\Pr[\exists a : T_a > 4n_a^+/p_a] \leq k \cdot \delta/(4k) = \delta/4$.

Total label complexity. Conditioned on all $T_a \leq 4n_a^+/p_a$:

$$n_{\text{Phase 1}} = \sum_{a=1}^k T_a \leq \sum_{a=1}^k \frac{4n_a^+}{p_a} = \frac{128}{\gamma^2} \ln \frac{8k}{\delta} \cdot \sum_{a=1}^k \frac{1}{p_a}.$$

Adding Phase 2: $n_{\text{Phase 2}} = O(d\epsilon^{-2} \ln(1/\delta))$. Total:

$$n = O\left(\frac{d}{\epsilon^2} \ln \frac{1}{\delta} + \sum_{a=1}^k \frac{\ln(k/\delta)}{\gamma^2 p_a}\right).$$

When $p_a = p_+$ for all groups: $n = O(d\epsilon^{-2} \ln \delta^{-1} + k(\gamma^2 p_+)^{-1} \ln(k/\delta))$.

The constant-factor doubling from sample splitting does not affect the asymptotic rate.

Overall success probability. Union bound over all failure events: $\delta/4$ (VC uniform convergence on S_{cstr} , Step A.1) + $\delta/4$ (feasibility of h^* , Part B) + $\delta/4$ (accuracy, Part C) + $\delta/4$ (termination, Part D) = δ .

This completes the proof for the EO case. □

B.6.3 EqOdds Case.

PROOF. The EqOdds case extends the EO proof by additionally requiring FPR estimation. Phase 1 collects both $2n_a^+$ positive and $2n_a^-$ negative instances per group (each split into constraint and validation halves), where:

$$n_a^- = \left\lceil \frac{32}{\gamma^2} \ln \frac{8k}{\delta} \right\rceil = n_a^+.$$

The negative instances are i.i.d. from $\mathcal{D}(X \mid Y=0, A=a)$, enabling unbiased FPR estimation. The same Hoeffding + union bound argument gives $|\text{FPR}_a(\hat{h}) - \text{FPR}_a(h)| \leq \gamma/8$ for all a , with an additional failure probability of $\delta/4$.

The expected Phase 1 cost per group is:

$$\mathbb{E}[T_a] = \frac{2n_a^+}{p_a} + \frac{2n_a^-}{1-p_a} = \frac{64}{\gamma^2} \ln \frac{8k}{\delta} \left(\frac{1}{p_a} + \frac{1}{1-p_a} \right),$$

yielding the stated complexity. The overall union bound accounts for TPR estimation, FPR estimation, accuracy, and termination events, each with failure probability at most $\delta/4$, for a total of at most δ . □

B.7 Proof of Corollary 1: Fairness Complexity Hierarchy

PROOF. The hierarchy $L_{\text{DP}} = 0 < L_{\text{EO}} = \Theta(k/(\gamma^2 p_+)) \leq L_{\text{EqOdds}} = \Theta(k/(\gamma^2 p_+ p_-))$ follows from combining the lower bounds (Theorem 1) and upper bounds (Theorem 3).

$L_{\text{DP}} = 0$. Lemma 2 shows DP verification requires only unlabeled data, so fairness adds zero label cost.

$L_{\text{EO}} = \Theta(k/(\gamma^2 p_+))$. Lower bound: Lemma 3 gives $\Omega(k \min(\epsilon, \gamma)/(\gamma^2 p_+))$; in the typical regime $\epsilon = \Omega(1)$, this is $\Omega(k/(\gamma^2 p_+))$. Under the excess-risk formulation (Remark 2), the full $\Omega(k/(\gamma^2 p_+))$ bound holds without restriction. Upper bound: Theorem 3 gives $O(k \ln(k/\delta)/(\gamma^2 p_+))$. Thus $L_{\text{EO}} = \Theta(k/(\gamma^2 p_+))$ up to $\ln(k/\delta)$ factors.

$L_{\text{EqOdds}} = \Theta(k/(\gamma^2 p_+ p_-))$. Lower bound: $\max(k/(\gamma^2 p_+), k/(\gamma^2 p_-)) \geq k/(2\gamma^2 p_+ p_-)$ by the AM-GM inequality $\max(a, b) \geq (a+b)/2$ applied to $a = 1/p_+$, $b = 1/p_-$:

$$\max\left(\frac{1}{p_+}, \frac{1}{p_-}\right) \geq \frac{1}{2} \left(\frac{1}{p_+} + \frac{1}{p_-} \right) = \frac{1}{2} \cdot \frac{p_+ + p_-}{p_+ p_-} = \frac{1}{2p_+ p_-}.$$

(The same $\min(\epsilon, \gamma)$ factor from Lemma 3 applies to both TPR and FPR lower bounds; see the discussion above for EO.) Upper bound: Theorem 3 gives $O(k \ln(k/\delta)/\gamma^2 \cdot (1/p_+ + 1/p_-)) = O(k \ln(k/\delta)/(\gamma^2 p_+ p_-))$.

Strict inequalities. $L_{\text{DP}} < L_{\text{EO}}$: Since $k/(\gamma^2 p_+) > 0$ for all $p_+ > 0$. ✓

$L_{\text{EO}} < L_{\text{EqOdds}}$ for all $p_+ \in (0, 1)$: The ratio $L_{\text{EqOdds}}/L_{\text{EO}} = 1/p_- = 1/(1-p_+) > 1$ whenever $p_+ > 0$. This is strict for all $p_+ \in (0, 1)$, not just $p_+ \neq 1/2$. □

B.8 Proof of Theorem 4: SUM-type Lower Bound

PROOF. The accuracy term $\Omega(d/\varepsilon^2)$ follows from Lemma 1. We prove the fairness term $\Omega(\sum_a 1/(\gamma^2 p_a))$ in three steps.

Step 1: Relabeling construction. Fix $h^* \in \mathcal{H}$ and a reference TPR value $\tau \in [1/4 + \gamma, 3/4 - \gamma]$. Define

$$q_a := \Pr[h^*(X)=1 \mid A=a], \quad r_a^+ := \Pr[Y=1 \mid h^*(X)=1, A=a], \quad r_a^- := \Pr[Y=1 \mid h^*(X)=0, A=a].$$

These satisfy $p_a = q_a r_a^+ + (1 - q_a) r_a^-$ and $\text{TPR}_a(h^*) = r_a^+ q_a / p_a$. For each group a and index $\theta_a \in \{-1, +1\}$, define the perturbation

$$r_a^+(\theta_a) = r_a^+ + \theta_a \Delta_a, \quad r_a^-(\theta_a) = r_a^- - \theta_a \Delta_a \cdot \frac{q_a}{1 - q_a},$$

where $\Delta_a := \gamma p_a / q_a$. Three properties hold:

(P1) *Marginal invariance:* $q_a r_a^+(\theta_a) + (1 - q_a) r_a^-(\theta_a) = p_a$ for both $\theta_a = \pm 1$. Verification: $q_a \cdot \theta_a \Delta_a + (1 - q_a) \cdot (-\theta_a \Delta_a \cdot q_a / (1 - q_a)) = \theta_a q_a \Delta_a - \theta_a q_a \Delta_a = 0$.

(P2) *Feature invariance:* Only $\Pr[Y \mid X, A=a]$ is modified, not $\Pr[X \mid A=a]$. Therefore unlabeled data carries zero information about θ_a .

(P3) *TPR separation:* $\text{TPR}_a(h^*; \theta_a=+1) - \text{TPR}_a(h^*; \theta_a=-1) = 2\Delta_a q_a / p_a = 2\gamma$.

Validity: For $\gamma < \min(\tau/2, (1-\tau)/2)$, the perturbed rates $r_a^\pm(\theta_a)$ remain in $(0, 1)$.

Step 2: Per-group information bound. Fix all groups $b \neq a$ and consider distinguishing $\theta_a \in \{-1, +1\}$ from labeled queries to group a . Each label yields observation $(Y, h^*(X))$. Because $\Pr[Y=1 \mid A=a] = p_a$ is the same under both hypotheses (by P1), the KL divergence decomposes as

$$D_{\text{KL}}(P_{-1} \| P_{+1}) = p_a \cdot D_{\text{KL}}(\text{Ber}(\text{TPR}_a^-) \| \text{Ber}(\text{TPR}_a^+)) + (1 - p_a) \cdot D_{\text{KL}}(\text{Ber}(\text{FPR}_a^-) \| \text{Ber}(\text{FPR}_a^+)). \quad (25)$$

TPR term. Since $\text{TPR}_a^\pm \in [\tau - \gamma, \tau + \gamma] \subseteq [1/4, 3/4]$, the Bernoulli KL bound (4) with $\delta = 2\gamma$ gives

$$D_{\text{KL}}(\text{Ber}(\text{TPR}_a^-) \| \text{Ber}(\text{TPR}_a^+)) \leq \frac{4\gamma^2}{\tau(1-\tau) - 4\gamma^2} =: C_1 \gamma^2. \quad (26)$$

FPR term. From the construction, the FPR change is $|\text{FPR}_a^+ - \text{FPR}_a^-| = 2\gamma p_a / (1 - p_a)$. Applying (4):

$$D_{\text{KL}}(\text{Ber}(\text{FPR}_a^-) \| \text{Ber}(\text{FPR}_a^+)) \leq C_2 \frac{\gamma^2 p_a^2}{(1 - p_a)^2}, \quad (27)$$

where C_2 depends on p and $\bar{p}$ through the bounds on $\text{FPR}_a^\pm$.

Total KL per query. Substituting (26) and (27) into (25):

$$D_{\text{KL}}(P_{-1} \| P_{+1}) \leq C_1 p_a \gamma^2 + C_2 \frac{p_a^2}{1 - p_a} \gamma^2 = p_a \gamma^2 \underbrace{\left(C_1 + \frac{C_2 p_a}{1 - p_a} \right)}_{\leq C_3 \text{ since } p_a \leq \bar{p} < 1} \leq C_3 p_a \gamma^2. \quad (28)$$

Sufficient statistic. The relabeling perturbs $\Pr[Y=1 \mid X, A=a]$ by a constant on each of $\{x : h^*(x)=1\}$ and $\{x : h^*(x)=0\}$, so $h^*(X)$ is a sufficient statistic for θ_a . Thus D_{KL} based on the full observation (X, Y) equals the D_{KL} based on $(h^*(X), Y)$ computed above.

Applying Le Cam's method. For adaptive queries, the chain rule (Lemma 4) gives total divergence $D_{\text{KL}} \leq \mathbb{E}[n_a] \cdot C_3 p_a \gamma^2$. By Pinsker's inequality, $\text{TV} \leq \sqrt{\mathbb{E}[n_a] \cdot C_3 p_a \gamma^2 / 2}$. Le Cam's bound requires $\text{TV} \geq 1/2$ for success probability $\geq 3/4$, giving

$$\mathbb{E}[n_a] \geq \frac{1}{2 C_3 p_a \gamma^2} =: \frac{c}{\gamma^2 p_a}. \quad (29)$$

Step 3: Summation over groups. Construct a composite instance $\theta = (\theta_1, \dots, \theta_k) \in \{-1, +1\}^k$ where each group is independently perturbed. *Cross-group independence:* observations from group b are identically distributed under $\theta_a=+1$ and $\theta_a=-1$ for $b \neq a$, because the perturbation for group a modifies only $\Pr[Y \mid X, A=a]$ and leaves $\Pr[X \mid A=a]$ unchanged (P2). Therefore the per-group lower bound (29) applies independently to every group:

$$\mathbb{E}[n] = \sum_{a=1}^k \mathbb{E}[n_a] \geq c \sum_{a=1}^k \frac{1}{\gamma^2 p_a}. \quad \square$$

C Extended Experimental Details

Table 4 summarizes the key properties of all datasets. The “Budget” column reports the total label budget allocated per experiment (across all methods), and “ p_+ ” is the empirical marginal positive rate.

Budget measurement. Throughout all experiments, “budget” refers to the total number of label queries (i.e., oracle calls for Y). Group memberships A and features X are freely observable from the unlabeled pool. For FAIRSTRATIFIED, the budget is $n_{\text{Phase 1}} + n_{\text{Phase 2}}$; for baselines, it is the cumulative number of queried labels.

Table 4: Dataset statistics. n : sample size after preprocessing, k : number of protected groups, d : feature count (before one-hot encoding), p_+ : marginal positive rate.

Dataset	n	k	d	p_+	Budget
Folktables	195,665	9	10	0.37	40K
COMPAS	6,167	6	7	0.45	4K
Adult	48,842	5	14	0.24	20K
Synthetic	500,000	varies	varies	varies	adaptive

C.1 Dataset Preprocessing

Folktables. We use the ACSIncome task from the 2018 American Community Survey via the folktables Python package (v0.0.12, $n = 195,665$ after removing rows with missing values). Features include age, education, work hours, occupation, and other economic indicators (10 continuous/ordinal features after one-hot encoding of categorical variables). The protected attribute is RAC1P (race), re-coded into $k=9$ groups: White, Black, American Indian, Alaska Native, Asian, Hawaiian/Pacific Islander, Other, Two Races, and Three+ Races. We apply StandardScaler (zero mean, unit variance) to all features. The dataset is split 80/20 stratified by both Y and A using `sklearn.model_selection.StratifiedShuffleSplit` with group-label pair (A, Y) as the stratification key. Binary label: income > 50K.

COMPAS. We use the ProPublica recidivism dataset (`compas-scores-two-years.csv`), following the preprocessing of Angwin et al. [5]: we filter to $-30 \leq \text{days_b_screening_arrest} \leq 30$ and $\text{is_recid} \neq -1$, yielding $n = 6,167$ instances. Protected groups ($k=6$) based on race: Caucasian, African-American, Hispanic, Asian, Native American, Other. Features ($d=7$): age (continuous), `priors\_count` (integer), `days\_in\_jail` (computed as `c\_jail\_out - c\_jail\_in`, continuous), `c\_charge\_degree` (binary: felony/misdemeanor), sex (binary), `age\_cat` (ordinal: < 25, 25–45, > 45), `score\_text` (ordinal: Low/Medium/High). Binary label: `two\_year\_recid`. Instances with missing values are dropped (<2%). Continuous features are standard-scaled.

Adult. We use the UCI Census Income dataset (`adult.data` and `adult.test` combined, $n = 48,842$). Protected groups ($k=5$) based on race: White, Black, Asian-Pac-Islander, Amer-Indian-Eskimo, Other. Features ($d=14$ before encoding): age (continuous), `workclass` (one-hot, 8 levels), `education-num` (ordinal), `marital-status` (one-hot, 7 levels), `occupation` (one-hot, 14 levels), `relationship` (one-hot, 6 levels), `capital-gain` (continuous), `capital-loss` (continuous), `hours-per-week` (continuous), `native-country` (binary: US/Other). Continuous features are standard-scaled; the effective dimensionality after one-hot encoding is 57. Binary label: income > 50K. The dataset is split 80/20 stratified by both Y and A .

Synthetic data. We generate data from a Gaussian mixture model with the following specification:

- *Feature distribution:* each group $a \in [k]$ has conditional distribution $X | A=a \sim \mathcal{N}(\mu_a, \sigma^2 I_d)$ with $\sigma = 0.5$. Group means $\mu_a \in \mathbb{R}^d$ are equally spaced on the unit sphere: $\mu_a = (\cos(2\pi a/k), \sin(2\pi a/k), 0, \dots, 0)$ for the first two coordinates, with the remaining $d-2$ coordinates set to zero. Group proportions are uniform: $\pi_a = 1/k$.
- *Label model:* $Y | X \sim \text{Ber}(\sigma_{\text{logistic}}(w^\top X + b))$ where $\sigma_{\text{logistic}}(z) = 1/(1 + e^{-z})$ is the logistic function, $w = (1, 0, \dots, 0) \in \mathbb{R}^d$ is the Bayes-optimal direction, and the intercept b is determined numerically (via bisection on 100K samples) to achieve the target marginal base rate $p_+ = \Pr[Y=1]$.
- *Label noise:* for experiments requiring non-realizable distributions, we replace the logit with $w^\top X + b + \eta$ where $\eta \sim \mathcal{N}(0, \sigma_\eta^2)$. In all reported experiments, $\sigma_\eta = 0$ (realizable setting) unless explicitly stated. The Bayes error under default parameters is <0.001, confirming approximate realizability.
- *Pool size:* fixed at 500,000 to avoid finite-pool effects. We verified that increasing to 1,000,000 does not change results.

C.2 Hyperparameter Selection

Logistic regression. All experiments use ℓ_2 -regularized logistic regression (`sklearn.linear_model.LogisticRegression`) with the following settings: regularization strength $C = 1.0$, solver `lbfgs`, maximum iterations `max_iter = 1000`, convergence tolerance `tol = 10^{-4}`, `class_weight=None` (no reweighting). We verified that $C \in \{0.01, 0.1, 1.0, 10.0, 100.0\}$ produces similar qualitative results (scaling slopes within $\pm 5\%$ across all datasets); $C = 1.0$ was chosen as it provides the best bias-variance trade-off measured by 5-fold cross-validation accuracy on the full labeled pool.

Constrained ERM. The constrained optimization $\min_{h: \widehat{\mathcal{F}}(h) \leq \gamma/2} \widehat{\text{err}}(h)$ is solved via the exponentiated gradient reduction of Agarwal et al. [3], implemented using the `fairlearn` library (v0.9.0) with the `ExponentiatedGradient` meta-algorithm. The inner convex program uses CVXPY (v1.4) with the ECOS solver. Convergence is declared when the duality gap satisfies $|\text{primal} - \text{dual}| < 10^{-4}$ or after 50 outer iterations (whichever comes first); all experiments converge within 20 iterations.

Batch active learning. Active methods (Active-Fair, FAL) use batch size $B = 10$ with a 50-sample warm-start drawn uniformly at random. The warm-start ensures the initial classifier has non-degenerate predictions before selection begins. We verified that $B \in \{1, 5, 10, 20\}$ and warm-start sizes in $\{20, 50, 100\}$ produce results within the standard error bands for all datasets.

C.3 Baseline Details

All baselines share the same classifier family (ℓ_2 -regularized logistic regression, $C = 1.0$), 80/20 stratified splits, random seeds ($s \in \{0, \dots, 9\}$), and evaluation protocol (test-set accuracy and fairness gap on the held-out 20%).

Passive-Fair. Draws labeled samples uniformly at random (standard passive learning). Fairness is enforced via post-hoc ROC threshold adjustment [48]: we reserve 20% of the labeled budget as a calibration set (disjoint from the training set used for classifier fitting and the held-out test set). For each group a , the classification threshold $\tau_a \in [0, 1]$ is chosen on this calibration set to equalize TPR across groups (for EO) or both TPR and FPR (for EqOdds). The threshold search uses 200 linearly spaced candidates per group, selecting the vector that minimizes EO (or EqOdds) on the calibration set. If no threshold vector achieves $\text{EO}(h) \leq \gamma$, the vector minimizing EO is returned.

Active-Fair. Uses uncertainty sampling with the same post-hoc fairness adjustment as Passive-Fair. The acquisition function selects the $B = 10$ instances with highest predictive entropy:

$$\alpha_{\text{unc}}(x) = H(\hat{p}(y | x)) = -\hat{p} \log \hat{p} - (1 - \hat{p}) \log (1 - \hat{p}),$$

where $\hat{p} = \hat{p}(Y=1 | x)$ is the current model's predicted probability and $H(\cdot)$ denotes the binary entropy function. The model is re-trained from scratch after each batch of B queries. To reduce computational cost, selection is performed over a random subsample of $\min(10,000, |\text{remaining pool}|)$ instances from the unlabeled pool.

FAL (Fair Active Learning). Follows Anahideh et al. [4]. The acquisition function combines uncertainty with a group-balancing term:

$$\alpha_{\text{FAL}}(x, a) = H(\hat{p}(y | x)) + \frac{\lambda}{n_a + 1},$$

where n_a is the number of labels already queried from group a and $\lambda = 0.5$ is the balancing coefficient (following the default in the original paper). We verified that $\lambda \in \{0.1, 0.5, 1.0, 2.0\}$ produces similar final accuracy and fairness gaps (within 1 standard error); $\lambda = 0.5$ was retained for consistency with the original implementation. The term $\lambda/(n_a + 1)$ upweights underrepresented groups. The same per-batch retraining and candidate subsampling as Active-Fair is used. Post-hoc threshold adjustment is applied identically to Passive-Fair.

FAIRSTRATIFIED (ours). Uses the two-phase design of Algorithm 2. Phase 1 draws instances from each group a sequentially, querying labels until n_a^+ positives are collected (using the calibrated stopping rule in Section C.6 for real data, or the exact theoretical rule for synthetic experiments). Phase 2 draws n_{train} i.i.d. labeled samples, trains a logistic regression model, and solves the constrained ERM $\min_{h: \widehat{\text{EO}}(h) \leq \gamma/2} \widehat{\text{err}}(h)$ via the exponentiated gradient reduction (Section C.2). Unlike the baselines, fairness is enforced *in-processing* via constrained optimization rather than post-hoc threshold adjustment. No per-batch model retraining is needed in Phase 1, which explains the lower computational cost (Table 8).

C.4 Label Savings Computation

For Table 3, *label savings* is defined as follows. Let n_{method} denote the smallest label budget at which method M first achieves the target operating point: test accuracy $\geq \tau_{\text{acc}}$ and $\text{EO} \leq \gamma$. Let n_{passive} be the same for Passive-Fair. Then:

$$\text{Saved} = 1 - \frac{n_{\text{method}}}{n_{\text{passive}}}.$$

If a method never reaches the target within the allocated budget, it receives 0% savings. The accuracy thresholds are set to 95% of the full-data accuracy for each dataset: Folktables $\tau_{\text{acc}} = 0.95 \times 82.5\% = 78.4\%$, COMPAS $\tau_{\text{acc}} = 0.95 \times 67.1\% = 63.7\%$, Adult $\tau_{\text{acc}} = 0.95 \times 85.2\% = 80.9\%$, where the full-data accuracies are computed by training logistic regression on the entire labeled pool.

C.5 Scaling Law Sweep Configurations

For Q1 (Fig. 1), we use baseline parameters $d=10$, $k=4$, $\gamma=0.05$, $p_+=0.25$, $\varepsilon=0.05$ and vary one parameter at a time while holding others fixed:

- $d \in \{5, 10, 20, 40\}$ (holding $k=4$, $\gamma=0.05$, $p_+=0.25$).
- $k \in \{2, 4, 8, 16\}$ (holding $d=10$, $\gamma=0.05$, $p_+=0.25$).
- $\gamma \in \{0.10, 0.05, 0.025, 0.0125\}$ (holding $d=10$, $k=4$, $p_+=0.25$).
- $p_+ \in \{0.50, 0.25, 0.10, 0.05\}$ (holding $d=10$, $k=4$, $\gamma=0.05$).

This yields $4 \times 4 = 16$ configurations total. For each, we run 10 seeds and record the label budget at which FAIRSTRATIFIED first achieves both $\text{err}(h) \leq \varepsilon$ and $\text{EO}(h) \leq \gamma$. Synthetic experiments use the exact theoretical constant ($32/\gamma^2$, not the calibrated $2/\gamma^2$) to validate the theoretical predictions directly.

C.6 Theoretical vs. Calibrated Constants

The theoretical stopping rule for Phase 1 requires $n_a^+ = \lceil (32/\gamma^2) \ln(8k/\delta) \rceil$ positives per group. The constant 32 arises from: (i) the factor of 2 in Hoeffding’s two-sided bound, (ii) the factor of $1/8^2 = 1/64$ from targeting precision $\gamma/8$ (not γ) to accommodate error-budget splitting across feasibility, fairness guarantee, and accuracy guarantee, and (iii) the $\ln(8k/\delta)$ accounts for a union bound over k groups with failure budget $\delta/4$ per event.

For real-data experiments, we use the calibrated rule $n_a^+ = \lceil (2/\gamma^2) \ln(2k/\delta) \rceil$, which corresponds to a Hoeffding bound targeting precision γ directly:

$$\Pr[|\widehat{\text{TPR}}_a - \text{TPR}_a| > \gamma] \leq 2 \exp(-2n_a^+ \gamma^2) = \frac{\delta}{k}.$$

The factor-of-16 gap between theoretical $(32/\gamma^2)$ and calibrated $(2/\gamma^2)$ constants does not affect the scaling behavior $\Theta(k/(\gamma^2 p_+))$, as validated on synthetic data using exact theoretical constants (Figure 1).

C.7 Statistical Testing

All reported results are means $\pm$ standard errors over 10 independent random seeds. Each seed determines a fresh 80/20 stratified split and a new random sequence for active query selection.

For the scaling law fit $n = c_1 d/\epsilon^2 + c_2 k/(\gamma^2 p_+)$, we use ordinary least squares on the 16 synthetic configurations (4 parameters $\times$ 4 values each). The $R^2 = 0.97$ is computed on these 16 data points. We also verified the fit is robust to leave-one-out cross-validation ($R_{\text{LOO}}^2 = 0.95$).

For the $1/p_+$ scaling of Figure 6, we fit $\log n = \beta_0 + \beta_1 \log p_+$ via OLS on the 8 data points with $p_+ \in \{0.01, 0.02, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$. The fitted slope $\hat{\beta}_1 = -0.97$ with $R^2 = 0.97$ confirms the theoretical $1/p_+$ prediction (expected slope = -1).

C.8 Extended Results: DP Comparison

Table 5 reports the performance of CONSTRAINEDERM-DP on the three benchmark datasets under DP with $\gamma = 0.10$.

Table 5: Performance under DP constraint ($\gamma=0.10$). CONSTRAINEDERM-DP achieves competitive accuracy with zero additional label cost for fairness; all label queries serve accuracy training.

Dataset	Method	Budget	Acc.(%)	DP	Unlabeled
Folktables	Passive-Fair	10K	78.9 $\pm$ 0.5	.087	–
	CONSTRAINED-DP	10K	81.7 $\pm$ 0.4	.068	15K
COMPAS	Passive-Fair	3K	65.5 $\pm$ 1.0	.090	–
	CONSTRAINED-DP	3K	67.8 $\pm$ 0.8	.071	8K
Adult	Passive-Fair	8K	84.1 $\pm$ 0.4	.085	–
	CONSTRAINED-DP	8K	85.5 $\pm$ 0.3	.063	12K

C.9 Extended Results: EqOdds Comparison

Table 6 extends Table 3 to the EqOdds setting, requiring both TPR and FPR equalization.

The EqOdds budgets are approximately $1.33\times$ the EO budgets for Folktables and Adult (where $p_+ \approx 0.25$, giving $1 + p_+/p_- \approx 1.33$), and only slightly larger for COMPAS ($p_+ \approx 0.45$, giving $1 + p_+/p_- \approx 1.82$, but COMPAS is already budget-constrained). Label savings are slightly lower under EqOdds (32–35% vs. 36–39% under EO) because the additional FPR estimation reduces the budget available for Phase 2 accuracy training.

C.10 Sensitivity to γ and δ

Table 7 examines how the choice of fairness tolerance γ and failure probability δ affects label complexity on synthetic data ($d=10$, $k=4$, $p_+=0.25$).

Varying γ . The theoretical Phase 1 cost scales as $1/\gamma^2$. Doubling γ (from 0.05 to 0.10) reduces Phase 1 cost by $4\times$, consistent with the quadratic dependence. However, the realized EO gap also increases proportionally: at $\gamma=0.20$, the EO gap is 0.15, which may be unacceptable in practice.

Varying δ . Phase 1 cost scales as $\ln(1/\delta)$. Reducing δ from 0.10 to 0.01 increases cost by a factor of $\ln(80k)/\ln(8k) \approx 1.5$, a modest overhead for substantially higher confidence.

Table 6: Performance under EqOdds constraint ($\gamma=0.10$). Phase 1 collects both positives and negatives per group, increasing label cost by a factor of $1 + p_+/p_-$ relative to EO.

Dataset	Method	Budget	Acc.(%)	EO	max Δ FPR	Saved
Folktables	Passive-Fair	55K	77.8 $\pm$ 0.7	.095	.098	–
	Active-Fair	55K	79.1 $\pm$ 0.6	.091	.094	18%
	FAL	55K	79.9 $\pm$ 0.5	.086	.088	24%
	FAIRSTRATIFIED	55K	81.5$\pm$0.4	.079	.082	35%
COMPAS	Passive-Fair	5K	63.8 $\pm$ 1.3	.098	.097	–
	Active-Fair	5K	64.9 $\pm$ 1.1	.094	.093	15%
	FAL	5K	65.6 $\pm$ 1.0	.090	.089	21%
	FAIRSTRATIFIED	5K	67.3$\pm$0.9	.084	.086	32%
Adult	Passive-Fair	28K	83.0 $\pm$ 0.5	.094	.096	–
	Active-Fair	28K	83.7 $\pm$ 0.5	.090	.092	17%
	FAL	28K	84.3 $\pm$ 0.4	.085	.087	23%
	FAIRSTRATIFIED	28K	85.2$\pm$0.3	.076	.079	34%

Table 7: Sensitivity to γ and δ on synthetic data ($d=10, k=4, p_+=0.25$). Phase 1 labels: $\sum_a n_a^+/p_a$. Realized EO: achieved EO gap at convergence.

γ	δ	n_a^+	Phase 1 labels	Realized EO	Acc.(%)
0.02	0.10	529,014	8.46M	0.015	87.2
0.05	0.10	84,642	1.35M	0.038	86.8
0.10	0.10	21,161	338K	0.072	86.5
0.20	0.10	5,290	84.6K	0.148	86.1
0.05	0.01	104,893	1.68M	0.036	86.9
0.05	0.05	91,038	1.46M	0.037	86.8
0.05	0.10	84,642	1.35M	0.038	86.8
0.05	0.20	78,247	1.25M	0.039	86.7

C.11 Computational Cost

Table 8 reports wall-clock times on a single NVIDIA A100 GPU (40GB) with 32-core AMD EPYC 7543 CPU, 256GB RAM, running Ubuntu 22.04. All timing is measured via `time.perf_counter()` in Python 3.10. FAIRSTRATIFIED’s overhead comes primarily from the constrained ERM step (exponentiated gradient reduction), not from Phase 1 sampling.

Table 8: Wall-clock time per seed (secs). “P1”: Phase 1 sampling, “P2”: Phase 2 training + constrained ERM, “Total”: end-to-end.

Method	Folktables	COMPAS	Adult
Passive-Fair	12s	2s	6s
Active-Fair	45s	8s	22s
FAL	52s	10s	26s
FAIRSTRATIFIED (P1)	8s	1s	4s
FAIRSTRATIFIED (P2)	18s	4s	10s
FAIRSTRATIFIED (Total)	26s	5s	14s

Notably, FAIRSTRATIFIED is *faster* than Active-Fair and FAL because Phase 1 uses simple random sampling (no model retraining per batch), while uncertainty-based methods retrain the classifier at each batch step.

C.12 Reproducibility

Software. Python 3.10.12, NumPy 1.25.2, scikit-learn 1.3.2, CVXPY 1.4.1, fairlearn 0.9.0, folktables 0.0.12, pandas 2.1.1, matplotlib 3.8.0. All libraries are installed via pip from PyPI.

Randomness. All random state is controlled via `numpy.random.seed(s)` for $s \in \{0, 1, \dots, 9\}$. This seed determines: (i) the 80/20 stratified train/test split, (ii) the active query sequence, (iii) the warm-start sample for active baselines, and (iv) the constrained ERM initialization.

Code. The complete source code, data preprocessing scripts, configuration files for all experiments are available for review and future research at <https://github.com/anonymous-ai-researcher/kdd2026>.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009